

# **Reproducibility in Artificial Intelligence Research in Embodied and Multi-Agent Autonomous Learning Systems: Voyager Case Study**

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## Abstract

This study investigates reproducibility challenges in embodied and multi-agent AI systems through a comprehensive case study of Voyager, an autonomous learning agent designed to navigate and progress in Minecraft's technology tree. While reproducibility is fundamental to scientific research, embodied AI systems present unique challenges due to their interactions with dynamic environments and complex inter-agent communications. Our reproduction attempts with the original Voyager implementation revealed a nuanced reproducibility landscape: early-game technological achievements (wooden, stone, and iron tools) were consistently reproducible across runs, while advanced achievements (diamond tools) exhibited substantial variability and environmental dependence. Most notably, we encountered a complete reproducibility failure in the inference module, achieving 0/9 successful task completions compared to the reported 100% success rate. Through detailed analysis of behavioral patterns and system interactions, we identified key reproducibility barriers including environmental variability, inter-agent communication issues, input ambiguity, and skill library overfitting. Our modifications targeting these issues improved advanced tool achievement rates from 33.3% to 83.3% and inference module success from 0% to 22%. These findings suggest that reproducibility in embodied AI requires considering not just final performance results but also behavioral consistency across learning trajectories. Additionally, we demonstrate how lifelong learning systems face unique challenges as knowledge integrity deteriorates over time, with errors compounding and creating divergent learning paths. Our work provides novel insights into the unique challenges at the intersection of embodied agents, multi-agent systems, and large language models, with implications for future research in autonomous learning systems.

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# 1. Introduction

The emergence of Voyager (Wei et al, 2023), an AI agent capable of open-ended learning in Minecraft, marked a significant advancement in autonomous learning systems. By combining large language models (LLMs) with automated curriculum learning and retrieval-augmented generation (RAG), Voyager demonstrated the potential for LLM-powered systems to autonomously acquire, refine, and execute complex skills in open-world environments. The system's reported ability to progress through Minecraft's technology tree while discovering novel solutions highlighted promising directions for autonomous agents in procedurally generated worlds.

This study presents a systematic investigation of Voyager's reproducibility, examining not only the replication of reported results but also the underlying factors that influence behavioral consistency and knowledge acquisition in complex AI systems. While previous studies have examined reproducibility issues in general AI and some aspects of embodied systems like robotics (Henderson et al., 2018), our work provides novel insights into the unique challenges that emerge at the intersection of virtual embodied agents, multi-agent systems, and large language models.

Our reproduction results reveal a complex picture. For basic technological milestones during the learning phase, we successfully reproduced the original study's findings, achieving 100% success rates for wooden, stone, and iron tools. However, even these successfully reproduced outcomes showed

greater timing variability than reported - while the original paper reported wooden tools being crafted within  $6 \pm 2$  iterations and iron tools within  $21 \pm 7$  iterations, our reproduction showed ranges of 2-10 iterations for wooden tools and 10-49 iterations for iron tools. For advanced milestones, we matched the reported 33.3% success rate for diamond tools (2 out of 6 runs), though notably these successes occurred only in a single world configuration. Most strikingly, the inference module showed a complete reproducibility gap - while the original paper reported 100% success rate across various crafting challenges, our reproduction achieved 0/9 successful task completions.

Beyond these quantitative results, our investigation examines the root causes of reproducibility challenges in complex AI systems of this type. Through detailed analysis of both successful and failed reproduction attempts, we discuss how the interaction between environmental variability, multi-agent communication, and knowledge acquisition processes affects the stability and reproducibility of system behavior. Our investigation revealed how these factors compound over time, creating distinct challenges for reproducing results in complex AI systems that combine embodied learning with large language models.

This paper presents a detailed examination of several key mechanisms affecting reproducibility. We investigate the relationship between behavioral consistency and result reproducibility, analyzing how the

stability of agent behavior affects our ability to reliably reproduce results. We explore how environmental variability shapes learning trajectories, and examine the mechanisms through which errors accumulate and propagate in lifelong learning systems. Through careful analysis of multi-agent communication patterns, we reveal how agent interactions impact system stability. Finally, we propose practical approaches to improving system robustness while maintaining learning capabilities.

Through this comprehensive analysis, we aim to contribute to the development of more robust and reliable autonomous learning systems. Our findings suggest that

future work in this field must expand beyond simply reproducing final outcomes to consider the stability of the entire learning process. This includes examining the cognitive aspects of how agents process and integrate new information, the behavioral patterns that emerge from environmental interactions, and the relational dynamics between multiple agents in the system.

## **2. Related work**

### **2.1 Reproducibility Challenges in Scientific Research and AI**

Scientific progress depends on the ability to verify and build upon previous findings, yet researchers across disciplines increasingly face challenges in reproducing published results. This section examines the reproducibility crisis affecting scientific research broadly, before exploring the unique complexities that artificial intelligence introduces to this fundamental scientific principle.

#### **2.1.1 The Reproducibility Crisis in Scientific Research**

Reproducibility and replicability are fundamental principles of scientific research, yet concerns about replicability have led to discussions of a "reproducibility

crisis" across multiple disciplines. A large-scale replication effort in psychology

found that while 97% of original studies reported statistically significant results, only 36% of replication attempts achieved the same threshold (Open Science Collaboration, 2015).

The terminology surrounding reproducibility has historically been a source of confusion. Plesser (2018) and the National Academies of Sciences, Engineering, and Medicine (2019) offer competing definitions, with the terms often used interchangeably. For this study, we approach reproducibility as the ability of an independent team to obtain the same results using the original dataset, code, and computational methods.

This reproducibility crisis is attributed to several factors, including publication bias,

poor methodological practices, inadequate statistical power, and lack of transparency in data and code sharing (Chen & Taqa, 2020). The academic incentive structure, which prioritizes novel findings over replication, creates a bias against publishing negative results, leading to questionable research practices (Ioannidis, 2005). Open science initiatives have emerged as a potential solution, encouraging the sharing of data, code, and methodologies to enhance transparency and facilitate independent verification (Munafò et al., 2017). As Ioannidis (2005) notably observed, "the hotter a scientific field, the less likely the research findings are to be true"—a particularly relevant concern for AI research given its current prominence and rapid pace of development.

### 2.1.2 Specific Reproducibility Challenges in AI Research

Artificial intelligence research introduces additional layers of complexity to reproducibility concerns. Unlike traditional scientific disciplines with relatively well-defined experimental procedures, AI research faces unique challenges due to its high computational demands, stochastic training procedures, proprietary datasets, and inconsistent evaluation metrics.

Several interrelated factors contribute to AI's specific reproducibility challenges. Henderson et al. (2018) demonstrated that deep learning models are highly sensitive to hyperparameter settings and random initialization, with slight variations in random seeds or batch ordering potentially leading to substantially different outcomes.

This non-determinism makes it difficult to determine whether reported improvements reflect genuine methodological advances or simply favorable parameter tuning.

Furthermore, Pineau et al. (2021) highlight how inconsistent evaluation protocols further undermine reproducibility, with researchers frequently using different preprocessing steps, undocumented training modifications, or custom evaluation metrics. This issue is particularly problematic in reinforcement learning and multi-agent systems. Additionally, many AI models rely on datasets that are either private, proprietary, or dynamic, making direct reproduction impossible.

Due to these issues inherent and unique to Artificial Intelligence research, it is useful to think about its reproducibility in a different way than in other well-established research fields. In an attempt to address the inherent complexity involved in reproducing AI research, Semmelrock et al. (2023) categorize AI reproducibility into three distinct levels:

- **Experiment Reproducibility:** Obtaining identical results when executing an AI method under identical conditions (same dataset, code, hyperparameters, and computing environment). Even this basic level can fail due to hardware differences, floating-point precision issues, or undocumented dependencies.
- **Data Reproducibility:** Successfully re-implementing an AI method independently while yielding similar

results on the same dataset, despite differences in programming languages or software versions.

- **Method Reproducibility:** Maintaining core findings when applying an AI method to different but related datasets or through alternative implementations—the true test of generalizability.

As will be discussed in subsequent chapters of this paper, these reproducibility challenges are further amplified in complex AI systems such as embodied agents and multi-agent architectures, where environmental variables, agent interactions, and emergent behaviors introduce additional layers of variability.

## 2.2 Embodied AI and Multi-Agent Systems

The study of **embodied artificial intelligence (AI)** focuses on agents that interact with their environments through perception and action, rather than merely processing abstract data. Unlike purely symbolic AI, which operates on pre-defined logical rules, **embodied AI** emphasizes the **physical and situated nature of intelligence**, proposing that cognition arises through continuous interactions between an agent and its surroundings (Paolo et al, 2024). Embodied agents can be categorized into **physical** (e.g., robots) and **virtual** (e.g., digital agents in simulations), both of which share common principles but face distinct challenges and affordances.

### 2.2.1 Embodiment in Artificial Intelligence

Embodied artificial intelligence focuses on agents that interact with their environments through perception and action, rather than merely processing abstract data. This approach, rooted in cognitive science and robotics, views intelligence as emergent from sensorimotor interactions with the environment, contrasting with classical AI approaches that abstract cognition from physical experience. Paolo et al. (2024) argue that embodiment enables situated learning and is "fundamental to achieving generalizable intelligence," where knowledge is acquired through experiences rather than preprogrammed rules.

#### 2.2.2 Virtual Embodied agents

Virtual embodied agents exist within simulated environments while having an observable presence that can sense surroundings and take actions. These agents can interact not only with their environment but also with other agents and human users. Rickel (2001) describes them as "autonomous, animated agents that support face-to-face interaction with people in [interactive virtual] environments," highlighting their interactive nature. Unlike static machine learning models, embodied agents engage in continuous perception-action loops, making their learning non-deterministic and context-dependent.

#### 2.2.3 Multi-agent Systems

Multi-agent systems (MAS) comprise multiple autonomous agents interacting

within shared environments through cooperative, competitive, or mixed interactions. According to Dorri et al. (2024), MAS exhibit three key characteristics: (1) Distributed Intelligence, where decentralized decision-making enables addressing complex problems beyond individual agent capabilities; (2) Emergent Behaviors arising from local interactions and autonomy rather than explicit programming; and (3) Dynamic Environments requiring adaptation to both environmental changes and other agents' actions. While this distribution of intelligence allows MAS to tackle more complex tasks, it simultaneously introduces unpredictability and coordination challenges.

#### 2.2.4 Special Problems of Reproducibility in Embodied and Multi-Agent AI Research

The challenge of reproducibility in artificial intelligence (AI) research is **exacerbated in embodied and multi-agent settings**, where the complexity of interactions between agents and their environments introduces **additional layers of variability** beyond those seen in traditional AI models. Embodied AI agents, particularly those operating in dynamic, interactive settings, must continuously **perceive, reason, and act** based on sensory data, often in ways that are **non-deterministic and difficult to control**. Similarly, **multi-agent AI** introduces **emergent behaviors and communication inconsistencies**, further complicating efforts to replicate results reliably.

This section highlights three **key factors** that contribute to reproducibility challenges in embodied and multi-agent AI: **environmental variability, inter-agent communication, and compounding effects**.

##### 2.2.1.1 Environmental Variability

One of the primary barriers to reproducibility in embodied AI is the inherent **variability of the environment** in which an agent operates. Unlike standard AI models trained on **static datasets**, embodied agents interact with **dynamic environments** that may exhibit **unpredictable changes** due to **sensor noise, physics engine differences, or external disturbances**. Even in simulated environments, subtle changes—such as lighting conditions, object placements, or physics engine updates—can lead to **significantly different agent behaviors**, making strict reproducibility difficult to achieve.

In real-world robotics, this issue is even more pronounced, as slight variations in **hardware calibration, battery levels, sensor drift, or actuator noise** can lead to **non-identical performance outcomes**, even when the same experiment is repeated under seemingly identical conditions. **Deitke et al. (2020)** address this challenge with **RoboTHOR**, a simulation-to-reality platform designed to standardize testing conditions for embodied agents by minimizing environmental discrepancies between simulated training and real-world execution. However, while such frameworks **reduce variability**, they **do not eliminate it**, and reproducibility in embodied AI

remains highly dependent on **fine-grained control of environmental parameters**.

### 2.2.1.2 Inter-Agent Communication and Coordination

Multi-agent AI systems introduce an additional layer of reproducibility complexity due to the necessity for communication and coordination between agents. In decentralized multi-agent systems, agents often develop **emergent communication strategies**, making it difficult to replicate the same coordination patterns across different experimental runs. Unlike single-agent reinforcement learning, where behavior is determined by a direct policy-action mapping, multi-agent systems introduce non-linear dependencies where one agent's decision can alter the entire system's trajectory.

This issue is further compounded by the increasing integration of Large Language Models (LLMs) in multi-agent communication. While LLMs enhance the expressivity and adaptability of AI agents, they also introduce inherent variability in responses, even under deterministic settings (e.g., temperature = 0). A key challenge here is that LLM-powered agents do not always produce identical outputs for the same input, as their reasoning and language generation may be influenced by not only random variance due to the probabilistic nature of LLM outputs, but also contextual nuances and previous interactions. **Liu et al. (2024)** highlight this issue in their study on human-robot interaction, where even minor variations in phrasing led to significantly different responses from the robot, affecting

both its decision-making and the user's perception of its consistency. This mirrors the challenges seen in multi-agent coordination, where emergent communication strategies evolve dynamically, making it difficult to ensure strict reproducibility across experimental runs.

If LLM-powered social robots struggle with maintaining stable and predictable communication patterns even in relatively controlled interactions with humans, this issue becomes even more pronounced in decentralized multi-agent systems, where multiple autonomous agents must synchronize their actions without a fixed communication protocol.

### 2.2.1.3 Compounding Effects of Multi-Agent and Embodied Variability

The combination of **environmental variability** inherent to embodied agents and **multi-agent interaction dynamics** creates a compounding effect, further exacerbating reproducibility issues. In multi-agent reinforcement learning (MARL), agent behaviors are not only influenced by the **environment** but also by **each other**, leading to **non-deterministic, emergent strategies**. A slight **perturbation** in the training environment or communication protocol can trigger a **cascade of behavioral adaptations**, meaning that even **small deviations in initial conditions** can result in **entirely different experimental outcomes**.

Because of this, reproducibility in embodied and multi-agent AI is a **major scientific challenge**, driven by the **inherent**

**variability of dynamic environments, the unpredictability of inter-agent communication, and the compounding effects of emergent interactions.** Unlike static AI models, which can be **re-run on the same dataset**, embodied and multi-agent systems operate in **non-deterministic, evolving conditions**, making replication of experimental results particularly difficult.

## 2.3 Voyager System: Architecture and Reproducibility Challenges

The Voyager system represents an intersection of virtual embodied agency and multi-agent systems, inheriting the reproducibility challenges inherent to both paradigms. As a sophisticated AI agent operating in Minecraft, Voyager exemplifies how environmental variability and inter-agent communication complexities can impact research reproducibility.

### 2.3.1 Environmental Context: Minecraft as a Testing Ground

The primary environment for Voyager is Minecraft—an open-ended virtual world where the agent perceives and manipulates its surroundings according to its internal decision-making processes. Minecraft worlds are procedurally generated based on a **world seed**, which is either a numeric or text value that initializes the generation algorithm, determining terrain features, biomes, resource distribution, and structures ([https://minecraft.fandom.com/wiki/Seed\\_\(level\\_generation\)](https://minecraft.fandom.com/wiki/Seed_(level_generation))). In principle, using identical seeds in the same Minecraft

version should create identical world layouts.

However, perfect environmental reproducibility remains elusive even with identical seeds. Several factors introduce variability:

- **Chunk generation ordering** is not always consistent between runs
- **Entity behavior** (animals, hostile mobs) varies due to randomized AI patterns
- **Weather and time cycles** can influence visibility and mob spawning
- **Physics interactions** may resolve differently in complex scenarios

These subtle environmental differences can significantly influence an agent's learning trajectory, especially in exploration-heavy tasks where chance encounters with resources or obstacles shape subsequent decisions.

### 2.3.2 Multi-Agent Architecture

While externally appearing as a single agent in Minecraft, Voyager internally operates as a **multi-agent system** with specialized components that communicate through LLM-driven interactions. This architecture distributes cognitive processing across several key agents:

- **Curriculum Agent:** Receives environmental data and learning history, then prioritizes and selects the next learning objectives
- **Action Agent:** Transforms task descriptions into executable code

- **Critic Agent:** Evaluates task execution success and provides feedback

Each agent operates with specific LLM-based instructions and processes different input formats. Critically, the outputs of one agent directly influence the inputs and decisions of others, creating interdependencies that can amplify small variations in processing.

### 2.3.3 Lifelong learning

An important feature of Voyager is its reliance on lifelong learning in the form of 1) Curriculum Agent using previous actions and experiences to determine new tasks, and 2) a growing skill library containing successful mined functions that managed to solve previous tasks.

These two distinct sources inform the agents, most notably Curriculum Agent and Action Agent, by providing them with useful information about tasks that have already been completed, tasks that are too difficult to complete, and successful ways of solving previous tasks in the form of code that are supposed to be used as a starting point for solving related more complex tasks.

### 2.3.4 System Modules and Operation

Voyager's functionality is divided into two primary modules:

**Learn Module:** This exploratory phase enables the agent to autonomously discover and master skills in the Minecraft environment. The bot progressively:

- Explores the world and identifies potential tasks

- Attempts to execute tasks through code generation
- Evaluates success/failure and refines approach
- Stores successful skills in a persistent skill library

A distinctive feature of Voyager's learn phase is its reliance on the Curriculum Agent, which determines the learning trajectory by selecting tasks that maximize exploration and technological advancement. The instructions for this agent are intentionally broad to enable open-ended learning, creating significant latitude for different task suggestions even when processing identical inputs. This design choice, while enabling creative problem-solving, introduces substantial variability in learning trajectories. The agent may follow dramatically different technological paths in different runs, making it highly susceptible to environmental influences.

**Inference Module:** This task-execution phase focuses on completing specific user-defined objectives rather than open-ended exploration. The inference module:

- Processes the target objective;
- Decomposes it into smaller subtasks;
- Retrieves relevant skills from the library developed during the learn phase;
- Executes these skills in the order of subtasks to complete the final objective.

The inference module's performance is heavily dependent on the quality, diversity, and generalizability of the skill library developed during the learning phase. If critical skills were not discovered during learning, the inference module may struggle to complete related tasks efficiently or at all.

### 2.3.5 Inherent Variability Sources

The Voyager system's complexity introduces multiple sources of variability that challenge reproducibility efforts:

1. **Environmental inputs:** Different world seeds or even subtle differences in the same seed create different resource distributions, terrain challenges, and exploration opportunities

2. **Path dependency:** Current decisions are influenced by the cumulative history of previous interactions
3. **LLM indeterminism:** Even with temperature settings at zero, large language models may produce slightly different outputs for identical prompts in different contexts
4. **Inter-agent communication variance:** Small differences in one agent's output can cascade through the system as other agents process and respond to that information

These variability sources create compounding effects where small initial differences can result in dramatically different behavioral trajectories and skill acquisition patterns across runs.

## 3. Research Objectives and Approach

This study was designed with a two-fold approach to investigating reproducibility challenges in embodied AI and multi-agent systems, using Voyager as a case study.

Our primary objective was to assess the reproducibility of the original Voyager system by conducting multiple runs under varying conditions. This direct reproduction effort aimed to quantify the consistency and reliability of the system's performance against the benchmarks reported in the original paper (Wei et al, 2023). By analyzing variations across multiple runs in different environments, we sought to identify specific factors contributing to

reproducibility challenges in embodied AI research.

Our secondary objective built upon these findings. After identifying the sources of variability and inconsistency in the original system, we developed targeted modifications to the system prompts and retrieval-augmented generation (RAG) mechanisms. These modifications were specifically designed to address the inherent variability in:

1. **Environmental interactions** - By making the agent's learning process more robust to environmental differences

2. **Inter-agent communication** - By improving the consistency of information exchange between the system's internal agents
3. **Goal-directed behavior** - By enhancing the clarity and focus of agent objectives, particularly in the Curriculum agent

This dual approach allowed us to not only document reproducibility challenges but also explore potential solutions. By comparing the performance of both the original and

modified systems across multiple runs, we can provide insights into how embodied AI and multi-agent systems might be designed for greater reproducibility while maintaining performance. This comparative analysis helps bridge the gap between identifying reproducibility problems and developing practical strategies to mitigate them.

The methodology, results, and discussion that follow are structured to reflect this two-phase approach, presenting findings from both the direct reproduction attempts and the modified system implementation.

## 4. Methods

### 4.1 Defining reproducibility

In this study, we approach reproducibility as the ability of an independent team to obtain the same or similar results using the original dataset, code, and computational methods. However, the nature of embodied AI research presents unique challenges to traditional reproducibility definitions.

While we utilized the computational methods described in the original Voyager paper, it is important to note that the open-sourced codebase from the official Voyager [GitHub repository](https://github.com/MineDojo/Voyager) (<https://github.com/MineDojo/Voyager>) required several fixes to become operational. Though these fixes were relatively minor and preserved the intended functionality, the code used in our experiments is not entirely identical to the repository version.

Furthermore, the originality of input data is understood as using the same Minecraft version and world configuration. Since no world seed was provided by the original Voyager team, recreating the exact world used in their simulations was impossible. Additionally, as discussed previously, even identical world seeds cannot guarantee completely identical learning trajectories due to the non-deterministic elements in Minecraft environments.

Therefore, our reproducibility goal was to consistently reach the same or reasonably similar benchmarks as reported in the original paper, using computational methods, code, and environmental conditions and data that approximate the original as closely as possible within the inherent constraints of embodied AI

research. The reproducibility type we aim for is most similar to the **Method reproducibility** discussed in section 2.1.2.

## 4.2 Experimental Design

### 4.2.1 Phase 1: Direct Reproduction

To assess the reproducibility of the original Voyager system, we conducted multiple runs with the following configuration:

- Minecraft version v1.19 (identical to the original implementation)
- LLM models identical to those referenced in the original whitepaper, being gpt-4 for Curriculum, Action and Critic agents, and gpt-3.5 for Q&A subsystem of the Curriculum Agent.
- Six different runs of the learn phase distributed across three different world seeds (two runs per world) to account for environmental variability
- Three separate runs of the inference module for three different tasks in a single world, each using a skill library obtained from one of the learning phases

This setup allowed us to evaluate both the consistency of performance within the same environment (intra-world variation) and across different environments (inter-world variation).

### 4.2.2 Phase 2: Modified System Implementation

Based on the results and observations from Phase 1, we developed a modified version of

the Voyager system with targeted improvements across three main categories:

#### System Prompt Enhancements:

- **Action Agent:** Restructured the prompt with clearer definitions of provided data, segmented information into logical wholes, and implemented stringent requirements for generated functions to either complete tasks successfully with clear feedback or fail with explanatory feedback.
- **Curriculum Agent:** Refined the prompt structure with explicit definitions of core objectives, added general instructions for task selection prioritization, and defined valid task categories to encourage more coherent progression.
- **Q&A System Prompts:** Revised question and answer generation prompts to focus on technology progression and purposeful exploration, eliminating random questions that serve no clear objective. Modified answer formats to ensure they contain all necessary information for effective function creation.

#### Retrieval-Augmented Generation (RAG) Implementations:

- **Craftable Items Recipe Retrieval:** Implemented a specialized embedding-based retrieval system using a dedicated vector database (recipes\_vector\_DB.json) that matches questions to relevant item

recipes through semantic similarity and appends recipe information to agent queries.

- **Inventory State Delta Tracking:** Modified the system to capture and provide inventory and environment state changes before and after task execution to the critic agent, improving its ability to accurately judge task success.

#### **Agent Architecture Modifications:**

- **Function Separation:** Split the Critic agent's dual functionality of task success assessment and code description writing into separate system prompts with specialized instructions for each function.

We tested this modified system using the same experimental configuration as Phase 1 (six learn phase runs across three different worlds and three inference module tests).

### **4.3 Benchmarks**

We have used three benchmarks used in the original Voyager, and an additional analytical approach that was developed by us. It is worth noting that the benchmarks reported in the original whitepaper are not clearly defined, so the benchmarks we are using are reconstructed from our understanding of the results that are presented in the whitepaper.

#### **4.3.1 Tech Tree Progression**

A set of metrics that measure the agent's advancement through Minecraft's technological progression system:

- **Tool achievement efficiency:** The number of iterations required to first obtain a tool of each material tier (wooden, stone, iron, diamond). This measures how quickly agents can progress through the game's technology tree, with lower numbers indicating faster progression.
- **Tool achievement rate:** The success rate in obtaining each tool tier across all runs, measured as both a ratio (e.g., 2/6 runs) and percentage. This indicates the consistency and reliability of an agent's ability to reach each technological milestone, with special attention to diamond tier as the most advanced achievement.
- **Item acquisition diversity and consistency:** Beyond core tool progression, we track the acquisition patterns of a comprehensive set of items across multiple runs. Items are categorized into three progression-based tiers:
  - **Early Game Items:** Fundamental resources and basic tools necessary for initial progression (wooden tools, stone tools, basic resources like cobblestone, and essential crafting stations).
  - **Mid Game Items:** Intermediate resources and equipment (iron tools and armor, buckets, shields) that enable more efficient resource gathering and greater environmental interaction.

- **Late Game Items:** Advanced resources and equipment (diamond tools, enchanting equipment) that represent sophisticated mastery of the game's mechanics.

This analysis reveals whether the agent demonstrates consistent priorities and acquisition patterns across different runs, providing deeper insight into reproducibility challenges at different stages of progression. It helps distinguish between agents that follow reliable learning trajectories versus those that achieve milestones through more variable or opportunistic paths.

## 5. Results

Our reproduction attempts revealed both successes and notable discrepancies compared to the original reported results. The following analysis quantifies these differences and examines their underlying causes.

### 5.1 Tech Tree Progression

#### 5.1.1 Reproducibility of the Original Implementation

Graph 5.1: Original graph depicting item acquisition over iterations

#### 4.3.2 Inference Module Task Success

A set of metrics that measures the agent's ability to complete a specific task:

- **Success rate:** How many runs of the inference module have managed to complete a task in the specified number of iterations (50, as in the original whitepaper).
- **Task decomposition comparison:** A comparison of subtasks that the inference module constructs in order to complete the final task.
- **Task progression comparison:** Comparison of completed subtasks.

The original Voyager paper presented results suggesting a smooth progression through Minecraft's technology tree, as depicted in Graph 5.1. This graph shows a continuous acquisition of items throughout the learning phase, with clear milestone achievements of wooden, stone, iron, and eventually diamond tools. Table 5.1 provided quantitative support, showing success rates and iteration ranges for obtaining these tool tiers across three runs.

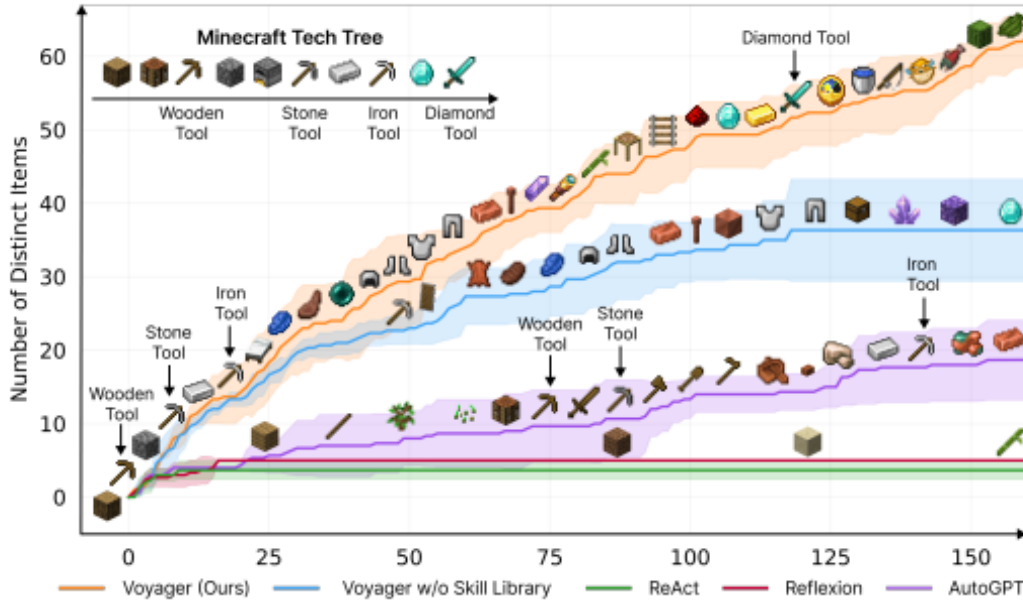


Table 5.1: Original table depicting the Voyager’s success rate in reaching tool tiers across different runs, along with the range of iterations it took in each to reach them

| Method                    | Wooden Tool   | Stone Tool    | Iron Tool       | Diamond Tool |
|---------------------------|---------------|---------------|-----------------|--------------|
| ReAct [29]                | N/A (0/3)     | N/A (0/3)     | N/A (0/3)       | N/A (0/3)    |
| Reflexion [30]            | N/A (0/3)     | N/A (0/3)     | N/A (0/3)       | N/A (0/3)    |
| AutoGPT [28]              | 92 ± 72 (3/3) | 94 ± 72 (3/3) | 135 ± 103 (3/3) | N/A (0/3)    |
| VOYAGER w/o Skill Library | 7 ± 2 (3/3)   | 9 ± 4 (3/3)   | 29 ± 11 (3/3)   | N/A (0/3)    |
| VOYAGER (Ours)            | 6 ± 2 (3/3)   | 11 ± 2 (3/3)  | 21 ± 7 (3/3)    | 102 (1/3)    |

Our replication attempt with six independent runs of the original Voyager implementation largely matches these results. As shown in Table 5.2, we found consistent achievement of early tool tiers (wooden, stone, and iron at 100% success rates), matching the original

paper's claims. For diamond tools—the most advanced tier—our results show a 33.3% success rate (2 out of 6 runs), which aligns with the original paper's 33.3% rate (1 out of 3 runs).

Table 5.2: Tool tier achievement rate

| Tool Tier | Source Prompts | Adjusted Prompts |
|-----------|----------------|------------------|
| Wooden    | 100% (6/6)     | 100% (6/6)       |
| Stone     | 100% (6/6)     | 100% (6/6)       |
| Iron      | 100% (6/6)     | 100% (6/6)       |
| Diamond   | 33.3% (2/6)    | 83.3% (5/6)      |

Table 5.3: Tool Progression Timing (iterations to achievement)

| Run | Version  | Wooden | Stone | Iron | Diamond |
|-----|----------|--------|-------|------|---------|
| 1   | Source   | 5      | 8     | 20   | -       |
| 2   | Source   | 6      | 11    | 39   | -       |
| 3   | Source   | 10     | 13    | 49   | -       |
| 4   | Source   | 7      | 10    | 18   | -       |
| 5   | Source   | 4      | 7     | 10   | 72      |
| 6   | Source   | 2      | 7     | 20   | 79      |
| 1   | Adjusted | 10     | 34    | 38   | 135     |
| 2   | Adjusted | 9      | 14    | 21   | -       |
| 3   | Adjusted | 11     | 14    | 23   | 104     |
| 4   | Adjusted | 16     | 22    | 30   | 61      |
| 5   | Adjusted | 4      | 12    | 14   | 157     |
| 6   | Adjusted | 7      | 11    | 15   | 76      |

However, an interesting detail that we observed during our reproduction attempts is that the two successful cases where the original implementation achieved diamond tools occurred in the same Minecraft world (with the same random seed). In the other

two worlds tested, the agent did not manage to reach this advanced tier.

Upon closer inspection of agents' behavior, it became apparent that while deciding on the next task, the Curriculum Agent often

made decisions based on the most immediately available resources within its range of vision: In an environment where the agent had diamond ores available within their range of vision earlier, the agent started thinking of diamonds and attempting to mine them much earlier, which led to crafting diamond tools. In contrast with this, in other cases the diamond ore was not in the immediate vicinity at any point during Voyager's exploration. In these runs, voyager often didn't start thinking about reaching diamond level tools, and would often proceed to complete other types of tasks that were not immediately important for tech progression.

This implies a degree of **environmental dependence** that was not immediately clear in the originally reported results. It suggests that environmental factors can play a significant role in determining the agent's ultimate success in reaching certain benchmarks: certain world configurations and resource distributions appear to encourage diamond tool acquisition while others prove to be challenging.

Another observation suggesting a greater variability than initially shown can be found in our data on **progression pacing**. The original whitepaper reported relatively consistent timing for early tool tier achievement across the three runs. As shown in Table 5.1, wooden tools were achieved within  $6 \pm 2$  iterations, stone tools within  $11 \pm 2$  iterations, and iron tools within  $21 \pm 7$  iterations. This narrow range, particularly

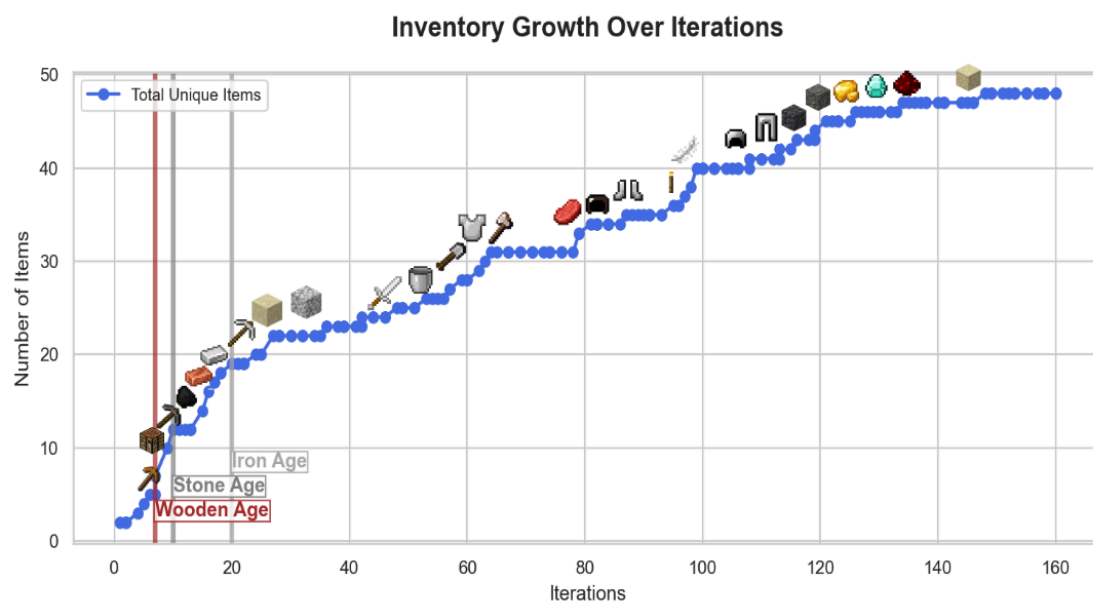
for wooden and stone tiers, suggested consistent early-game progression pacing.

Our reproduction results with the original implementation (Table 5.3) show greater variability across six runs: wooden tools required 2-10 iterations, stone tools 7-13 iterations, and iron tools 10-49 iterations. This expanded range, especially for iron tier, indicates that the original paper's evaluation may have understated the actual **pacing variability** inherent in the system. The increased variance in our data reveals that even when the agent consistently achieves a particular tool tier, the timing of these achievements is **less predictable** than suggested by the original study.

This difference in observed progression pacing is another sign pointing towards behavior that is not entirely consistent across runs. It also implies that with increasing complexity (the variability in the amount of iterations it took to reach iron tools is the greatest), the consistency of behavior decreases- similarly to how reaching diamond tools was the most inconsistently reached tool tier.

The graph 5.1 suggests that Voyager follows a smooth learning curve, acquiring a range of relatively **useful items** along the way. In order to track Voyager's consistency in acquiring useful items during the learning process, we have looked at its acquisition rates of a set of early, mid, and late game items over runs.

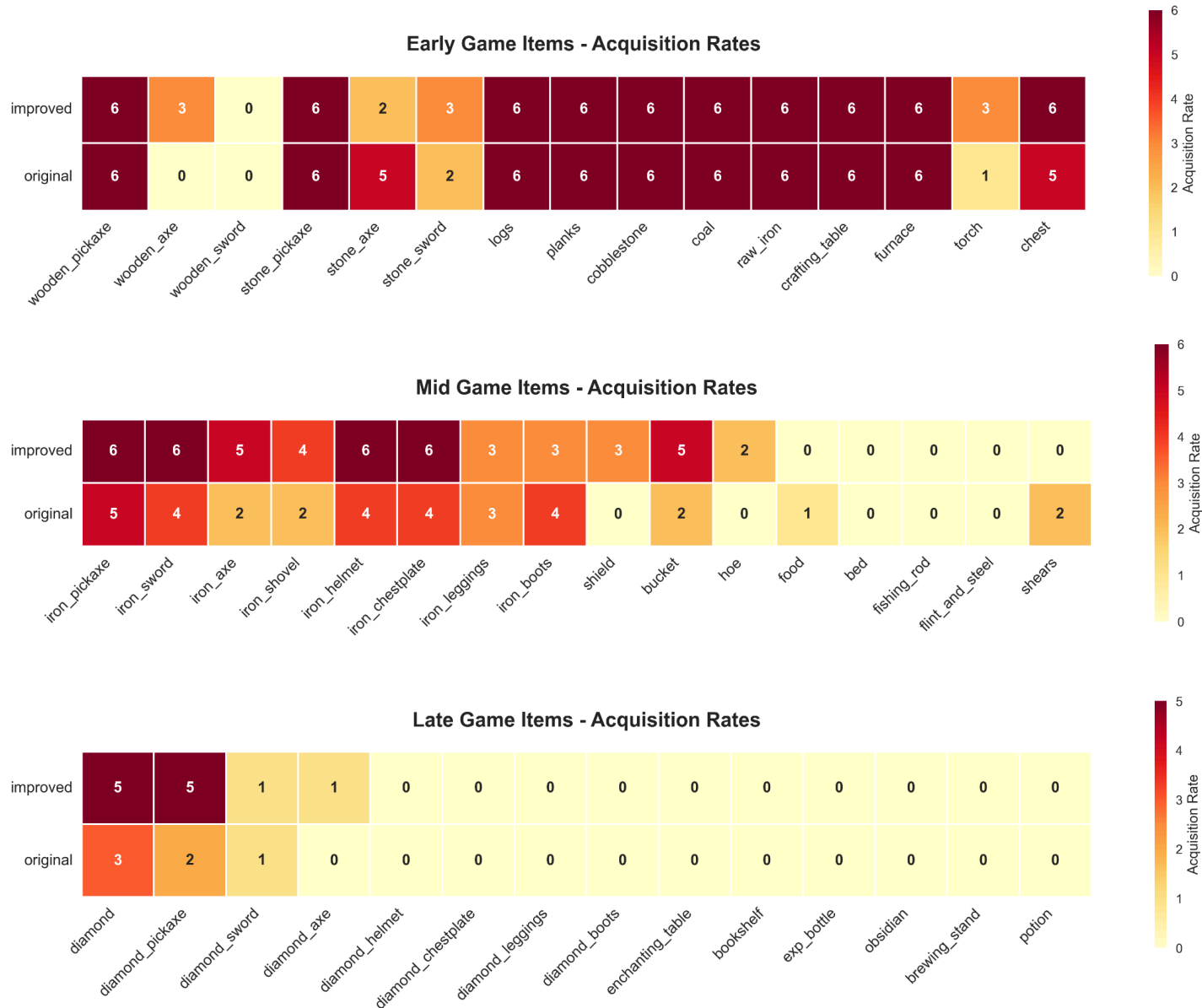
Graph 5.2: An example of item acquisition over iteration during reproduction attempts



The heatmaps shown in graph 5.3 show the results. The original implementation shows **high consistency** in acquiring fundamental **early-game resources** (wooden\_pickaxe, stone\_pickaxe, cobblestone, coal, raw\_iron, crafting\_table, and furnace all at 6/6 or 5/6).

However, **mid-game items** (Image 2) show **moderate variability**, with acquisition rates of iron tools ranging from 2/6 to 5/6. **Late-game items** demonstrate the **greatest variability**, with diamond acquisition at only 3/6 runs and diamond pickaxes at 2/6.

Graph 5.3: Acquisition rates of useful early, mid and late game useful item acquisition rates over separate runs



This data seems to reinforce the idea that the consistency in successful completion of tasks deteriorates in the later stages of the learning process. The more complex tasks whose completion is contingent on a longer chain of dependencies (completing previous tasks, navigating the environment, etc), the less predictability and consistency there is in their completion. Early-game decisions involve simpler environmental interactions with fewer dependencies, while late-game achievements require longer sequences of successful actions across more complex environments.

This observed behavioral pattern of decreasing consistency seems to be compatible with the discussion about specific challenges associated with reproducibility in embodied agents and multi-agent systems.

While behavioral consistency and result reproducibility are not synonymous, in this case, the behavioral consistency is directly related to result reproducibility: if reaching complex technological benchmarks is influenced by the environmental configuration, then the reproducibility of late benchmark achievements becomes inherently variable. Similarly, if there are growing LLM generated hallucinations and inter-agent miscommunications that are being integrated into Voyager’s learning process, distorting it as complexity grows, this also has a destabilizing effect on benchmark achievements.

In the Discussion section, we will pay more attention to the observed causes of these

behavioral distortions and their effects on consistency and reproducibility.

### 5.1.2 Improved implementation

Our **modified implementation** addressed some of the challenges identified in the original system through targeted enhancements to system prompts and retrieval-augmented generation mechanisms. The results demonstrate improvements in behavioral consistency and reproducibility across all metrics.

The diamond tool achievement rate increased from 33.3% to **83.3%** (5 out of 6 runs), presenting a significant improvement in behavioral stability and reproducibility.

Interestingly, while the original implementation only achieved diamond tools in a single world environment, our improved implementation successfully reached diamond tier in **all three tested worlds**. This improvement demonstrates that our modifications reduced the agent’s dependence on specific favorable world configurations, making it **more robust to environmental variability**. By enabling consistent achievement across different procedurally generated worlds, our modifications addressed the fundamental environmental dependency that undermined reproducibility in the original implementation.

This increase in consistent behavior is visible in item acquisition patterns (graph 5.3) as well. For **early-game items**, while both implementations show strong fundamental resource acquisition, our improved implementation demonstrates

more consistent acquisition of supporting items like wooden axes (3/6 vs. 0/6) and torches (3/6 vs. 1/6). These items, while not critical for advancement, facilitate more effective exploration and resource gathering.

The **mid-game** heatmap reveals more substantial improvements. Our modified implementation shows notably higher consistency in acquiring the complete set of iron equipment, with all core iron tools and armor showing acquisition rates of 4/6 or better, compared to much more variable rates in the original implementation. This suggests our modifications improved the agent's systematic prioritization and progression rather than focusing narrowly on core advancement items.

**Late-game items** show the most dramatic reproducibility improvements, with diamond

and diamond pickaxe acquisition increasing from 3/6 and 2/6 to 5/6 for both items, as well as managing to gain other diamond level tools in two other runs. With that said, however, the improvements did not have an effect on reaching other listed late game items.

## 5.2 Inference Module Task Success

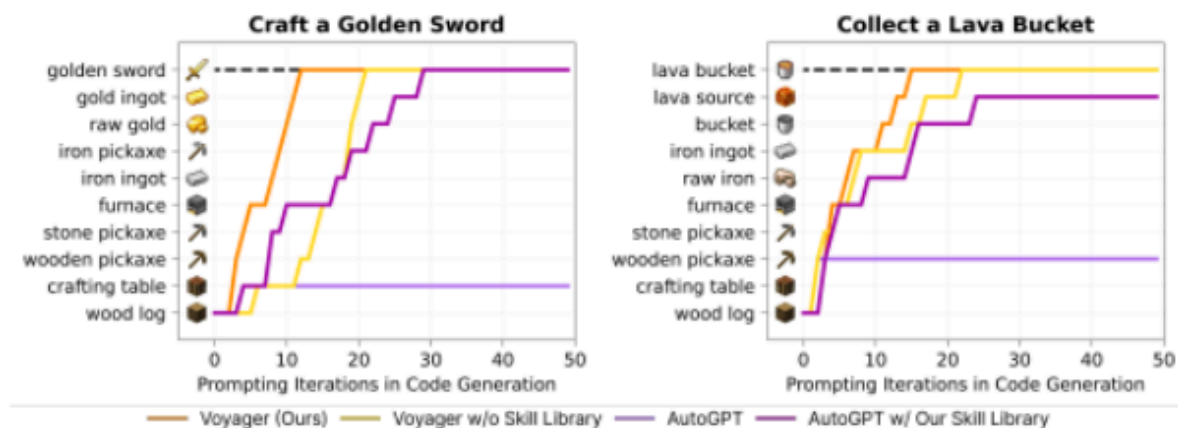
### 5.2.1 Original Paper Results

In the original Voyager paper, the authors evaluated the agent's **inference module** through a **zero-shot generalization test**, where Voyager was tasked with completing unseen tasks using its learned skill library. As shown in Table 2, the original implementation reported substantial success in completing various crafting challenges.

Table 5.4: Zero-shot generalization to unseen tasks.

Fractions indicate the number of successful trials out of three total attempts. 0/3 means the method fails to solve the task within the maximal prompting iterations (50). Numbers are prompting iterations averaged over three trials. The fewer iterations, the more efficient the method.

| Method                            | Diamond Pickaxe  | Golden Sword     | Lava Bucket      | Compass          |
|-----------------------------------|------------------|------------------|------------------|------------------|
| ReAct [29]                        | N/A (0/3)        | N/A (0/3)        | N/A (0/3)        | N/A (0/3)        |
| Reflexion [30]                    | N/A (0/3)        | N/A (0/3)        | N/A (0/3)        | N/A (0/3)        |
| AutoGPT [28]                      | N/A (0/3)        | N/A (0/3)        | N/A (0/3)        | N/A (0/3)        |
| AutoGPT [28] w/ Our Skill Library | 39 (1/3)         | 30 (1/3)         | N/A (0/3)        | 30 (2/3)         |
| VOYAGER w/o Skill Library         | 36 (2/3)         | 30 $\pm$ 9 (3/3) | 27 $\pm$ 9 (3/3) | 26 $\pm$ 3 (3/3) |
| VOYAGER (Ours)                    | 19 $\pm$ 3 (3/3) | 18 $\pm$ 7 (3/3) | 21 $\pm$ 5 (3/3) | 18 $\pm$ 2 (3/3) |



Graph 5.4: Zero-shot generalization to unseen tasks.

We visualize the intermediate progress of each method on two tasks. See Appendix, Sec. B.4.3 for the other two tasks. We do not plot ReAct and Reflexion since they do not make any meaningful progress.

For tasks like crafting a diamond pickaxe, the paper reported a **3/3 success rate** for Voyager with an average of 19 $\pm$ 3 iterations to completion. Similar success rates were reported for other complex tasks: golden sword (3/3, 18 $\pm$ 7 iterations), lava bucket (3/3, 21 $\pm$ 5 iterations), and compass (3/3, 18 $\pm$ 2 iterations). The paper also presented **progression curves** showing Voyager's acquisition of **intermediate items** on the path to task completion, illustrating a relatively **efficient progression** from basic to advanced resources.

The key findings about the inference module's capabilities in the original paper were:

1. **High success rates (100%)** for complex crafting tasks
2. Efficient task completion within a **reasonable number of iterations** (18-21 iterations)
3. **Logical progression** through prerequisite items
4. Ability to **leverage previously learned skills** for new challenges

These results suggested that Voyager could **reliably transfer knowledge** from its **learning phase** to complete **specific goal-directed tasks**, even those not explicitly encountered during training. This generalization capability is presented as a significant strength of the system.

### 5.2.2 Reproduction Results

Our reproduction study tested the inference module on three specific tasks: crafting a diamond sword, crafting a golden hoe, and crafting a lava bucket. To assess reproducibility, we conducted three runs for each task using the original implementation.

Table 5.5: Inference Task Success Rates Comparison Table

| Task                 | Original Implementation | Improved Implementation |
|----------------------|-------------------------|-------------------------|
| <b>Diamond Sword</b> | 0/3 (0%)                | 0/3 (0%)                |
| <b>Golden Hoe</b>    | 0/3 (0%)                | 2/3 (67%)               |
| <b>Lava Bucket</b>   | 0/3 (0%)                | 0/3 (0%)                |
| <b>Overall</b>       | <b>0/9 (0%)</b>         | <b>2/9 (22%)</b>        |

Our reproduction results with the original implementation diverged substantially from the reported performance in the paper. Across all three tasks and all runs (0/9 total), the original implementation failed to complete any task within the 50-iteration limit. This stark contrast with the original paper's reported 100% success rate represents a significant reproducibility failure.

### Task Decomposition Analysis

To understand the underlying causes of these failures, we analyzed how the original system decomposed each task into subtasks. Task decomposition quality directly impacts execution success, as improper sequencing of activities can create unsolvable situations.

Table 5.6: Task Decomposition Comparison Table

| Crafting Task        | Original Version                                                                                                                                                                                                                                                                                                                        | Improved Version                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Diamond Sword</b> | <ol style="list-style-type: none"> <li>1. Mine 2 diamond</li> <li>2. Mine 3 cobblestone</li> <li>3. Craft 1 wooden pickaxe</li> <li>4. Mine 3 iron ore</li> <li>5. Smelt 3 iron ore</li> <li>6. Craft 1 iron pickaxe</li> <li>7. Craft 1 diamond sword</li> </ol>                                                                       | <ol style="list-style-type: none"> <li>1. Mine 1 oak_log</li> <li>2. Craft 4 wooden_planks</li> <li>3. Craft 1 wooden_pickaxe</li> <li>4. Mine 3 stone</li> <li>5. Craft 1 furnace</li> <li>6. Craft 3 cobblestone</li> <li>7. Craft 1 stone_pickaxe</li> <li>8. Mine 3 iron_ore</li> <li>9. Smelt 3 iron_ore</li> <li>10. Craft 3 iron_ingot</li> <li>11. Craft 1 iron_pickaxe</li> <li>12. Mine 2 diamond</li> <li>13. Craft 1 stick</li> <li>14. Craft 1 diamond_sword</li> </ol> |
| <b>Golden Hoe</b>    | <ol style="list-style-type: none"> <li>1. Mine 2 gold ore</li> <li>2. Mine 2 sticks</li> <li>3. Smelt 2 gold ore</li> <li>4. Craft 1 golden hoe</li> </ol>                                                                                                                                                                              | <ol style="list-style-type: none"> <li>1. Mine 3 wood</li> <li>2. Craft 1 crafting table</li> <li>3. Craft 1 wooden pickaxe</li> <li>4. Mine 3 stone</li> <li>5. Craft 1 stone pickaxe</li> <li>6. Mine 1 iron ore</li> <li>7. Craft 1 furnace</li> <li>8. Mine 1 more wood for fuel</li> <li>9. Smelt 1 iron ore</li> <li>10. Craft 1 iron pickaxe</li> <li>11. Mine 1 gold ore</li> <li>12. Craft 2 sticks</li> <li>13. Craft 1 golden hoe</li> </ol>                              |
| <b>Lava Bucket</b>   | <ol style="list-style-type: none"> <li>1. Mine 3 deepslate</li> <li>2. Craft 1 wooden pickaxe</li> <li>3. Equip wooden pickaxe</li> <li>4. Mine 3 deepslate_iron_ore</li> <li>5. Craft 1 furnace</li> <li>6. Smelt 3 iron ore</li> <li>7. Craft 1 iron pickaxe</li> <li>8. Equip iron pickaxe</li> <li>9. Find 1 lava source</li> </ol> | <ol style="list-style-type: none"> <li>1. Mine 5 wood</li> <li>2. Craft 1 crafting table</li> <li>3. Craft 1 wooden_pickaxe</li> <li>4. Mine 3 stone</li> <li>5. Craft 1 stone_pickaxe</li> <li>6. Mine 3 iron_ore</li> <li>7. Craft 3 iron_ingot</li> <li>8. Craft 1 bucket</li> <li>9. Find lava</li> </ol>                                                                                                                                                                        |

|  |                                                    |  |
|--|----------------------------------------------------|--|
|  | 10. Craft 1 bucket<br>11. Collect 1 lava in bucket |  |
|--|----------------------------------------------------|--|

Analysis of the decomposition strategies reveals several critical issues in the original implementation:

1. **Logical dependency violations:** The original implementation consistently produced flawed task plans that violated technological dependencies. For example, when tasked with crafting a diamond sword, it attempted to "Mine 2 diamond" before establishing the prerequisite tool chain necessary to mine diamonds. For the golden hoe task, the agent planned to mine gold ore without first establishing iron tools needed to mine gold. These planning failures represent fundamental logical errors in the agent's reasoning.
2. **Incomplete prerequisite understanding:** The original system

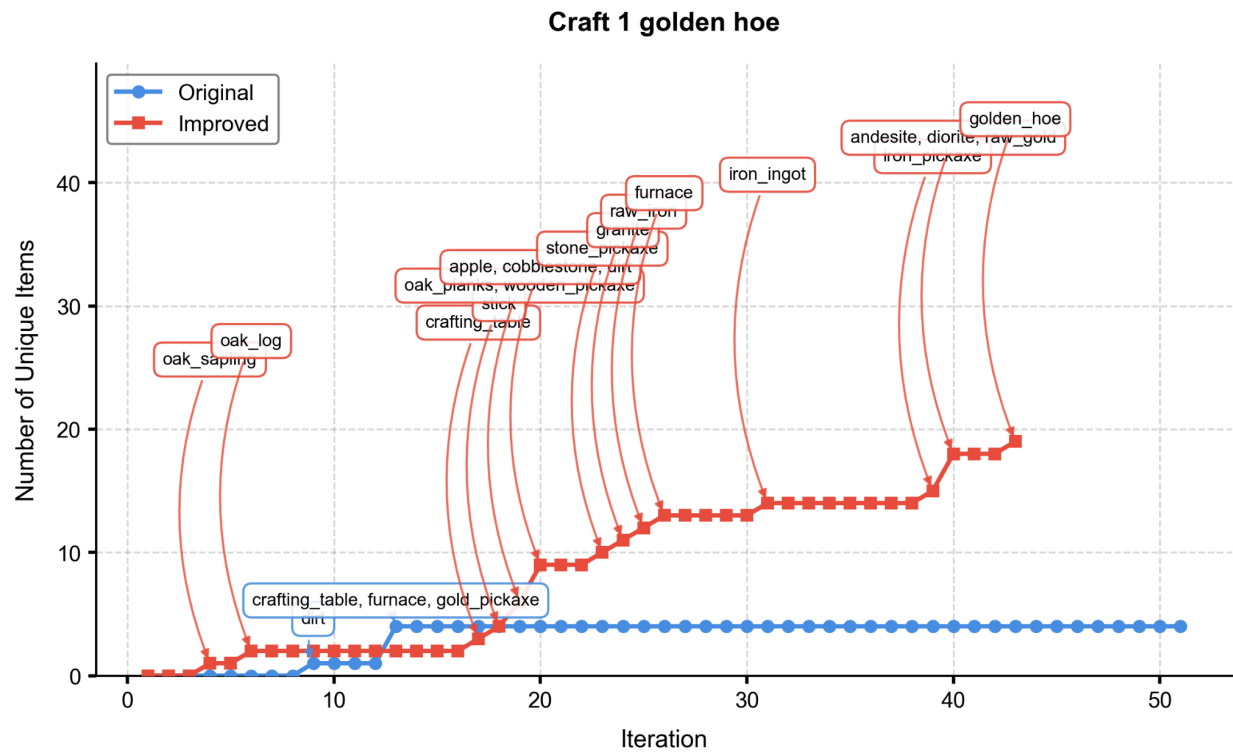
often omitted crucial intermediate steps, such as creating a crafting table before attempting crafting operations or obtaining wooden planks before crafting sticks.

3. **Simplified but impractical plans:** While the original system's plans were shorter (typically 4-11 steps compared to 9-14 steps in our improved version), this brevity came at the cost of feasibility. The plans lacked necessary preparation steps that would make execution possible in the Minecraft environment.

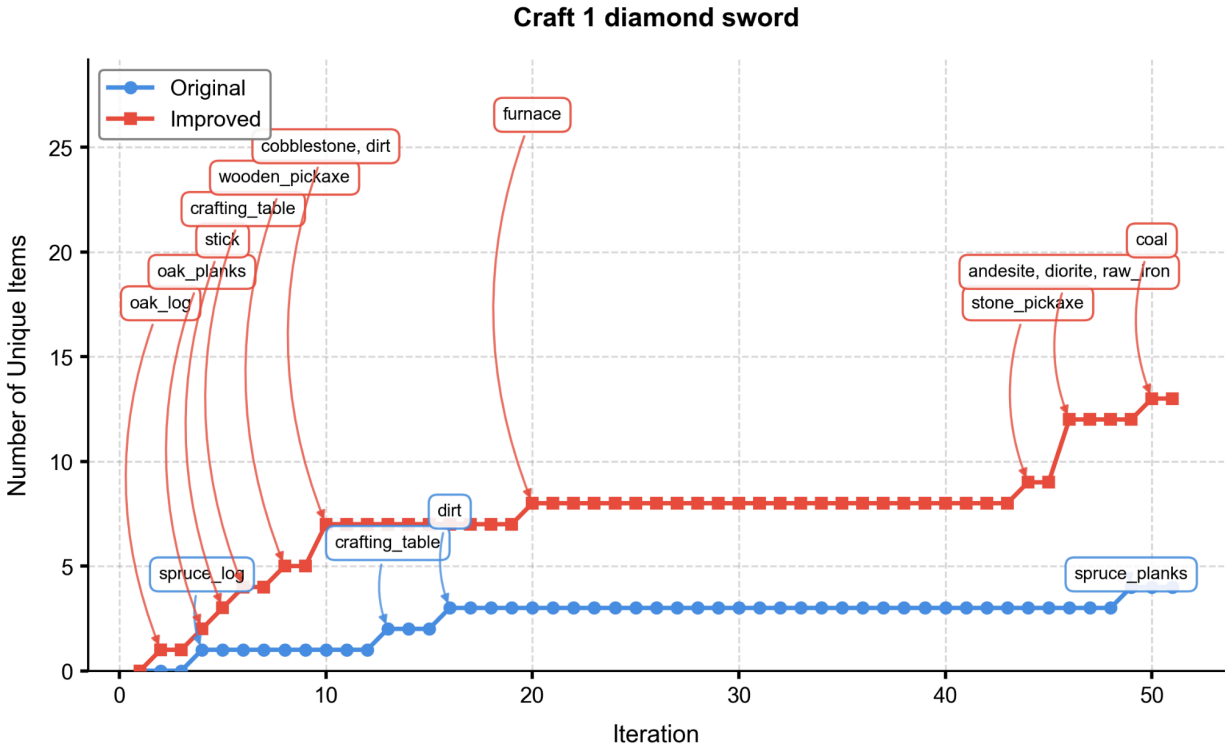
### Progression Analysis

The task progression curves provide visual evidence of how these decomposition issues affected execution performance over time.

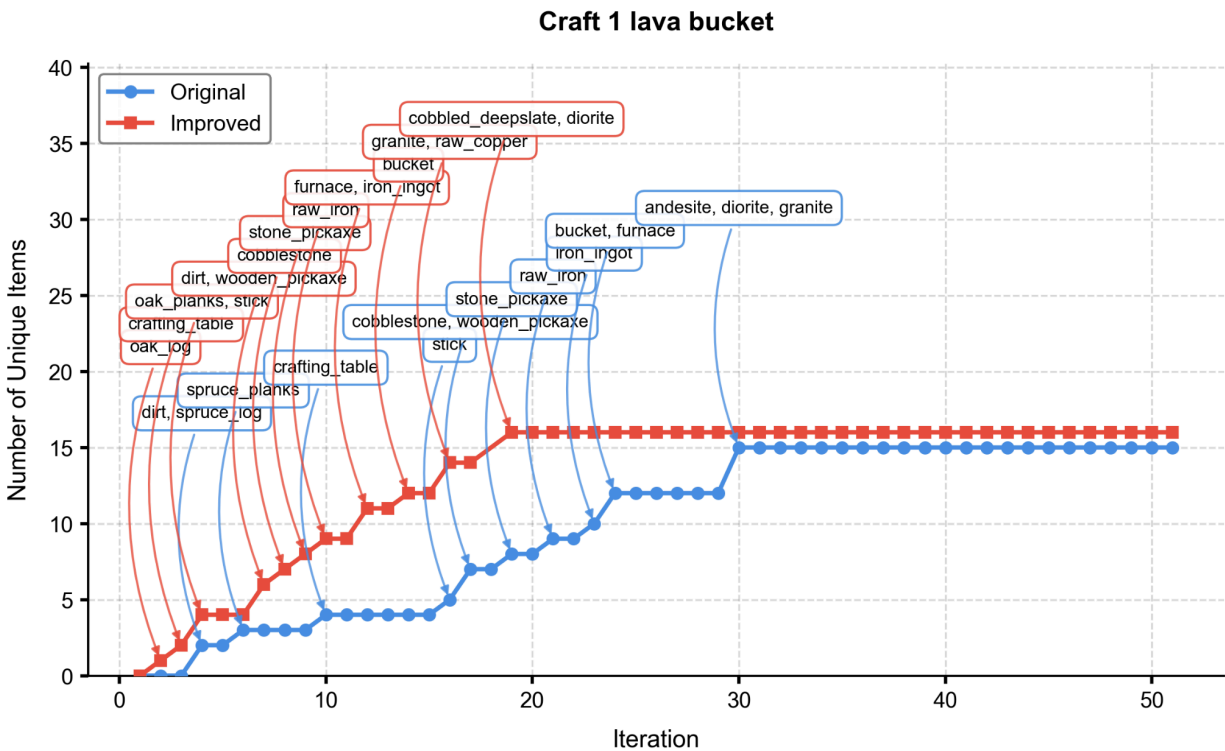
Graph 5.5: Task job inventory progression for original and improved prompt for ‘Craft 1 golden hoe’ task



Graph 5.6: Task job inventory progression for original and improved prompts for ‘Craft 1 diamond sword’ task



Graph 5.7: Task job inventory progression for original and improved prompts for 'Craft 1 lava bucket' task



Analysis of the task progression curves (blue lines in Graphs 5.5-5.7) reveals consistent patterns that illuminate important reproducibility challenges:

1. **Early plateaus:** Across all three tasks, the original implementation shows minimal initial progress (acquiring 4-5 items) before reaching extended plateaus where no new items are acquired. These plateaus typically occur after acquiring basic resources but before establishing mid-tier capabilities, suggesting the agent becomes stuck attempting impossible tasks without prerequisites.
2. **Execution stagnation:** In many cases, the progression curves show that the agent became completely stuck at certain technological levels, unable to advance despite continuing execution for dozens of iterations.
3. **Consistent failure pattern:** The consistency of failure across all nine runs (three per task) suggests that the performance issues are systematic rather than the result of random variation.

These consistent patterns of failure across all tasks suggest that the performance issues are systemic rather than random, pointing to fundamental limitations in how the original system handles planning and execution in embodied environments.

The failure to achieve any success and thus even partially reproduce the reported results

raises the question of what could be the cause of this. The flawed task decomposition that was already discussed is the most obvious reason for the lack of success.

However, we have also observed that many of the mineflayer functions generated during the learning phase and stored in the skill library, are too optimized for the specific circumstances in which they were generated, which limits their generalizability to similar tasks in similar situations. This is especially true for functions developed for complex tasks, which tended to be the most overfit ones.

### 5.2.3 Improved implementation

Our modified implementation addressed the task decomposition logic. The improvements yielded certain improvements in task completion, with our modified system achieving a 22% overall success rate (2/9 runs) compared to 0% for the original implementation. Most notably, the golden hoe task was successfully completed in 2/3 runs, demonstrating an improved performance for this particular challenge.

As shown in Graphs 5.5-5.7 (red lines), the qualitative improvements include:

1. **Logical task decomposition:** The improved implementation consistently generates appropriate task sequences that respect technological dependencies. For example, when crafting a diamond sword, it properly sequences from

wood gathering through stone tools to iron tools before attempting to mine diamonds. This logical structuring addresses the fundamental planning flaws in the original implementation.

2. **Consistent early progression:** Across all three tasks, the improved implementation shows rapid and consistent acquisition of fundamental resources (wood, crafting table, wooden tools) within the first 5-10 iterations. This reliable early-game progression provides a stable foundation for more advanced tasks.
3. **Later plateaus:** While the improved implementation still exhibits plateaus, these typically occur much later in the execution process (after 15-20 items), suggesting the agent reaches more complex challenges before encountering difficulties.
4. **Improved task handling:** While still falling short of the original paper's reported 100% success rates, our improved implementation demonstrates meaningful progress toward reproducing the claimed capabilities.

The successful runs for the golden hoe task are particularly informative. In these cases, the agent successfully followed its logical task decomposition from basic resources through to gold acquisition and tool crafting. This success, contrasted with the continued challenges in diamond sword and lava bucket tasks (both tasks that are more complex), suggests that task complexity and environmental factors interact to determine reproducibility outcomes.

These improvements demonstrate that addressing planning issues of the inference module in the original system can substantially enhance reproducibility of complex goal-directed behaviors. The consistency of early progression and logical task decomposition across all test cases suggests that our modifications successfully addressed some of the key reproducibility challenges inherent in the original system.

However, even with these improvements, perfect reproducibility remains elusive. The modified system still encountered execution difficulties in two of the three tasks. While improving task decomposition and planning helps the agent have a reasonable path forward and increases how likely it is to complete the task, there are still other issues that the current version of Voyager cannot handle. For example, when looking for lava or mining at lower block levels searching for diamonds, it is not uncommon for the agent to accidentally mine into lava, resulting in its death. At the same time, many of the advanced skills from the learning phase that are stored in the skill library are overoptimized for the specific situation in which they were generated, and this limitation still poses a significant hurdle to successful task completion.

Lastly, since the reproducibility failure is quite severe for the inference module, and even our improved results do not manage to approach the originally reported performance, the possibility of an undisclosed error in the published code is not out of the question.

## 6. Discussion

Our attempts to reproduce the behavior and results reported in the original Voyager whitepaper have helped us gain a better insight into the issues that make this type of system vulnerable to inconsistent and unpredictable behaviors, which has an effect on the overall reproducibility of reported results.

While our number of sample runs is insufficiently large to guarantee any statistically significant verdict on whether Voyager results can be said to be reproducible or not, the value of this paper lies in the plethora of important insights about inherent cognitive, behavioral and interactional vulnerabilities of Voyager as a system. These observed vulnerabilities, reflected in the reported results, are

In the following section, we take a look at the overall results; we map out the performance bottlenecks that we encountered, as well as their relation to each of the aspects of the system; and we map out ways in which our improvements helped mitigate some of the issues.

### 6.1 Reproducibility Assessment of Voyager

Our study revealed a contrast between the learning and inference modules of Voyager. For the learning phase, we largely succeeded in reproducing the high-level outcomes reported in the original paper. We matched the success rates for tool tier achievements (100% for wooden, stone, and iron tools; 33.3% for diamond tools), though we

observed greater timing variability than reported in the original work. While the paper reported wooden tools being crafted within  $6 \pm 2$  iterations, we observed ranges from 2-10 iterations. For iron tools, the original paper claimed  $21 \pm 7$  iterations, but we observed anywhere from 10-49 iterations. This suggests the original three runs may not have captured the full range of system variability.

In contrast, we completely failed to reproduce the results of the inference module. Despite following the methodology described in the paper, our reproduction achieved a 0% success rate across all tasks compared to the reported 100% success rate. This discrepancy suggests either undocumented implementation details, model variance, or bugs in the provided open sourced code.

Beyond simple reproducibility of reported results, our multiple runs revealed that the system's consistency exists on a continuum influenced by task complexity and environmental factors. We observed a clear pattern where reproducibility decreased as task complexity increased—early-game achievements showed high consistency, mid-game achievements showed moderate consistency, and late-game achievements showed low consistency. Additionally, diamond tool achievement only occurred in a single world configuration out of the possible three, suggesting a potential environmental dependency, although it could also be the product of random

probability, considering our small sample size.

Connecting to the reproducibility levels discussed in 2.1.2, it can be noted that we can reasonably only talk about a partial **Method reproducibility** when discussing Voyager.

**Experiment reproducibility**, which requires reaching identical results by using identical dataset, code, hyperparameters, and computational environment, was not possible to test for two reasons.

First, the provided open sourced code had errors that had to be fixed in order for us to be able to run Voyager. While we tried to fix the code without disturbing the underlying logic, it is a statement of fact that some pieces of code had been altered and that we cannot say with full certainty that the code is identical to the code used when the original whitepaper was written.

Second, the identical data that was used at the time of writing the whitepaper was not available to us. The original world seed was not provided in the open sourced material, so we weren't able to use the identical world configuration. Additionally, even if we did have the original world seed, considering the persisting elements of randomness in chunk generation, in-game mob behavior, as well as emergent behaviors of embodied agents, it is unlikely that an identical input data is possible even with the same world seed.

This is also the reason why testing for **data reproducibility**, which- although it allows for different implementation details- requires identical data, is not possible.

However, Voyager manages to achieve partial **Method reproducibility**, meaning it can reach similar results on a different implementation and with different data. This level of reproducibility seems to be the most relevant for embodied systems, since it acknowledges that the data and implementation details may vary across the runs.

Reaching this level of reproducibility means that the Voyager method has elements of generalizability when applied in differing environments, especially in the early phases.

However, in the later phases of the game progression, with increased complexity and prolonged learning time, the reproducibility decreases and the behavior becomes increasingly unpredictable.

## 6.2 Causes of Reproducibility Problems

There are several interconnected factors that contributed to the reproducibility challenges we encountered.

**Environmental variability** played a significant role in shaping agent behavior. Different resource distributions across world seeds affected the agent's reasoning about next tasks, while random encounters with hazards could terminate promising paths. These factors created strong path dependencies, where initial exploration patterns determined later technological progression possibilities. Voyager's tendency to be influenced by different environments often led to divergent learning trajectories, some of which moved away from

developing the high-level tools required to reach the later technological benchmarks.

**Inter-agent communication issues** undermined system consistency across multiple components. The Curriculum Agent, operating with vague directives like "explore and make Voyager the best minecraft player in the world," produced divergent interpretations across runs. Meanwhile, the Critic Agent's limited visibility into previous environment states prevented accurate assessment of task execution outcomes. The Action Agent faced a different problem: it generated code functions but lacked mechanisms to receive detailed execution feedback about why functions failed in specific environments, making systematic debugging and improvement impossible.

**Input ambiguity** further exacerbated these issues. The original implementation suffered from system prompts that provided insufficient definitional clarity, allowing LLMs to develop significantly different interpretations of the same input data on various runs. As an example, inventory data formatted as "Inventory: 0/31" led the system to incorrectly conclude the inventory was full (0 slots available) rather than empty (0 slots used out of 31).

**System design flaws** are an additional cause of these problems. Specifically, skills stored during the learning phase often contained contextually overfit solutions that couldn't be effectively transferred to new situations. The overfitting of solutions is likely caused by the design of the system that relies on immediate environmental feedback and error

messages in order to optimize the task-completing functions, leading to functions that are over-optimized for the specific context at hand.

These issues created powerful **compounding effects** where small initial variations became amplified through successive agent interactions, creating a "butterfly effect" in system behavior. As the system accumulated experiences, early divergences in learning paths became increasingly difficult to reconcile. Voyager, as an agent system that relies on previous experiences and lifelong learning to tackle new problems, was particularly vulnerable to these compounding errors. We identified **two distinct ways** errors become integrated into the agent's knowledge, corrupting its ability to adequately reason about new tasks, solve existing problems, or transfer skills:

**LLM context data contagion:** Each agent receives information regarding the surrounding world, previous behavior, completed or failed tasks, code used in previous iterations, and error messages emitted during execution. Critically, this input data is the product of behavior during previous iterations. Any hallucinations, agent miscommunications, or wrong conclusions become part of the next iteration's context, often with detrimental effects. For example, when a task fails multiple times due to hallucinations and gets classified as "too difficult," it permanently enters a list that the curriculum agent avoids. If this task is strategically important for game progression, this error becomes a permanent feature of the flawed knowledge base, adversely affecting future behavior.

These seemingly isolated failure points throughout the system produce long-term effects by contaminating the learning process, with more errors accumulating as iterations progress.

**Skill library overfitting.** Functions created during learning often become overfit to specific situations rather than serving their intended purpose as generalizable solutions. This overfitting occurs when functions, initially inadequate, are iteratively improved based on specific error feedback. Instead of creating genuinely generic solutions, the system adapts functions to specific circumstances (e.g., lack of materials), making them overfit to particular situations. A concrete example illustrates this problem: a single learning run might produce multiple functions for mining cobblestone, each solving the same task but overfit to different circumstances (no stone blocks nearby, lacking a pickaxe). This damages the skill library's utility in two critical ways: First, during the learning phase, when a new task is encountered and vector queries are executed, the skill library delivers multiple divergent overfit functions that all solve the same task. This both pushes other potentially relevant functions out of the limited list of returned functions and introduces redundant code that obscures which parts are actually useful for the new task. Second, during the inference phase, applying these context-specific functions in new environments proves extremely difficult. The functions, having been optimized for particular environmental conditions during learning, fail to generalize properly when transferred to new contexts with different conditions, explaining the

system's near-complete failure to accomplish inference tasks in our reproduction study.

### 6.3 Addressing the issues

Our modifications to the original system attempted to address the identified issues by: 1) increasing the clarity of input data, 2) minimizing inter-agent miscommunication, and 3) enhancing goal-oriented behavior to counter emergent behavior that was overly dependent on immediate surroundings. These interventions aimed to improve the learning process by reducing systemic vulnerabilities and, consequently, stabilize Voyager's internal reasoning and outward behavior.

It's worth noting that some system design issues identified as causes of learning problems were not addressed, as they fell outside our planned intervention scope. Specifically, the iterative function improvement mechanism using environmental feedback that caused skill library overfitting was not directly improved.

During the **learning phase**, these interventions greatly increased consistency and stability, leading to more predictable achievement of technological milestones—diamond tools success rate increased from 33.3% to 83.3%, with noticeable increases in mid-game item acquisition and mild improvements in late-game items. This reflects improvements in two critical areas:

**Improved learning process.** Increased quality of input processing, inter-agent

communication and better distribution of tasks among the agents results in a learning process that is much more resilient to the progressive incorporation of errors into the agent’s experience and behavior.

**Improved goal-oriented behavior.** The increased emphasis and clarification of the system’s purpose and goals introduced to the Curriculum Agent had a large impact on mitigating the observed behavioral and cognitive dependence on immediate surroundings to the detriment of pursuing the more distant goals that require forward-thinking and planning. With this change, new tasks are always evaluated both in comparison to tasks already finished, the immediate environment, as well as their usefulness and relevance for finishing future tasks and unlocking more advanced skills. This enhances the robustness of the system behavior in that its decisions are driven towards higher technological achievement at every iteration, while considering the past achievements and current state, which contributes to a more consistent behavior that is less ‘impulsive’ learning curve that develops into directions that are too influenced by immediate surroundings and increasingly distant from the desired benchmarks.

For the **inference phase**, our interventions yielded more modest improvements—task success rate increased from 0% to 22%, with improvements specific to the golden hoe task. This modest gain can largely be attributed to improvements in task decomposition and planning. All tasks demonstrated improved logical sequencing of subtasks, showing more consistent and

meaningful progress before hitting plateaus. However, complex task completion remained difficult due to the overfit functions from the skill library and challenges in applying them to new contexts.

While our interventions succeeded to improve the consistency and stability of agent’s behavior and cognition, there are still problems that are affecting the learning process and inference processes in potentially disruptive ways:

**Function adaptation mechanism.** We did not modify the original iterative mechanism that creates overfit functions. This limitation partly explains the limited improvements to the inference module—while better task decomposition logic pushed agents in the right direction, any thorough improvement would require addressing the fundamental mechanism of function creation and storage to build a more generalizable skill library.

**Lack of hazard mitigation.** An important limitation affecting both learning and inference processes is Voyager’s inability to efficiently avoid hazards such as lava, drowning, or hostile mobs. In practice, we observed Voyager often digging directly into lava while mining or reacting too slowly to drowning and mob attacks, resulting in death. These deaths disrupt the learning process and, in worst cases, create loops where Voyager repeatedly takes the same fatal path toward solving a task.

**Potential reduced exploration.** The improvements aimed at increasing goal-directed behavior might inadvertently

limit Voyager's open exploration capabilities. While these changes increase consistency and reproducibility of technological benchmarks, they may discourage exploration paths not connected to tech progression. Given that both technological progression and exploration are stated goals of Voyager, this potentially impacts its exploratory aspect.

## **6.4 Theoretical Implications**

Our findings have several important theoretical implications for understanding reproducibility in embodied AI and multi-agent systems.

### **6.4.1 Behavioral Consistency as a Prerequisite for Reproducibility**

In complex LLM-powered systems with multi-agent architectures that operate in embodied settings, behavioral consistency and result reproducibility are intrinsically linked. Traditional machine learning models can achieve reproducibility through fixed data and controlled parameters, but embodied agents face a fundamental challenge: they must maintain consistent behavioral patterns to achieve truly reproducible outcomes.

Our reproduction experiments with systems like Voyager revealed a critical insight: while we successfully reproduced the reported statistical results for the initial learning phase, this reproducibility heavily depends on the underlying processes governing system behavior. These behavioral processes, however, proved remarkably fragile. We observed decreasing

predictability and consistency as both learning duration extended and system complexity increased.

This finding points to a deeper issue in embodied AI reproducibility. Behavioral variability inherently increases with task complexity and learning duration, creating fundamental challenges for reproducing advanced capabilities. Two systems may achieve identical benchmark outcomes while following wildly different behavioral patterns—raising the question of whether they truly demonstrate reproducible intelligence.

Our research suggests that reproducibility assessment for embodied AI systems must extend beyond simple outcome metrics to include comprehensive behavioral analysis.

### **6.4.2 Knowledge Integrity in Lifelong Learning Systems**

Systems that rely on lifelong learning face unique reproducibility challenges related to knowledge integrity. Our findings demonstrate that what is learned at each stage influences all subsequent learning, creating path dependencies that can either facilitate or undermine reproducibility. When learning processes incorporate errors, hallucinations, or overspecialized knowledge, these deficiencies compound over time, creating divergent learning trajectories.

This suggests that reproducible lifelong learning requires mechanisms to maintain knowledge integrity throughout the learning process. Rather than focusing solely on final performance metrics, researchers should

evaluate the quality, consistency, and transferability of accumulated knowledge.

### **6.4.3 Balancing Direction and Exploration**

Agents operating in embodied environments that rely on lifelong learning will become influenced by said environment if they lack stronger internal direction. While introducing more direction in the agent's instructions will stabilize the behavior and make it more driven towards completing the specified goals, it may simultaneously discourage agents from pursuing divergent paths, and in this way may be at odds with its exploratory abilities.

Our observations suggest a potential tension between reproducibility and open-ended exploration. When enhancing goal-directed behavior, we observed improved consistency and reproducibility of technological progression, but this appeared to potentially constrain the system's ability to discover novel, unexpected solutions. This suggests that reproducibility and creative exploration may represent competing objectives in embodied AI design.

This apparent tension reflects a broader theoretical question in artificial intelligence: can systems designed for creative, open-ended exploration also maintain high levels of predictability and reproducibility? Our observations indicate that improvements in reproducibility may come with reduced behavioral diversity and exploratory potential. This observation connects with existing theoretical work on the exploration-exploitation dilemma (Berger-Tal et al, 2014; Yan et al, 2024) in

reinforcement learning, where maximum reproducibility might theoretically reduce the beneficial variability that enables new discoveries.

In practical terms, designers of embodied AI systems may need to carefully balance the degree of internal direction provided to agents based on their intended purpose. Systems prioritizing consistent, reproducible outcomes might benefit from stronger goal-directed constraints, while those aimed at novel discovery and adaptation might require greater freedom from rigid directives, accepting the associated unpredictability as a necessary condition for innovation.

### **6.5 Limitations**

Due to the small sample size, this study is unable to give any statistically significant statements about Voyager's behavior. Instead, it is intended to be a deep dive exploring the behavioral and cognitive patterns observed during our reproduction attempts and abstracting the crucial observations that are useful for developing more stable and reproducible embodied AI systems.

For a more statistically robust study, the number of 1) test runs and 2) world seeds would need to be significantly increased.

### **6.6 Future research**

While this study is designed to offer insight into the vulnerabilities faced by embodied learning agents that influence their ability to efficiently learn and complete tasks in a consistent manner, the number of runs we did is insufficient to grant any statistically

significant conclusions. However, these qualitative findings suggest several potentially useful research directions.

### **6.6.1 Quantitative reproducibility assessment**

Building on our qualitative findings, future research should conduct larger-scale reproducibility studies with sufficient runs to establish statistical significance. These studies should 1) quantify the extent and progression of behavioral variability across different task complexities; 2) evaluate the reproducibility impact of different system components through controlled ablation studies; and 3) develop standardized metrics for assessing behavioral consistency in addition to outcome reproducibility.

### **6.6.2 Generalizable skill acquisition**

The skill library overfitting problem identified in our study deserves particular attention. Seeking to develop and test alternative mechanisms of generating generalizable functions during the learning process in embodied environments is one of the potential venues of further research.

Addressing the overfitting that is caused by the current function adaptation mechanism needs to develop an alternative architecture that will govern the function development process in a way that emphasizes the modularity and generalizability of solutions.

One possible solution is already partially available in the existing system in the form of task decomposition used in the inference module. Applying task decomposition to every new task during the learning process,

combined with checks of already existing functions, could encourage a step-by-step approach to solving problems and developing modular functions that each solve a single problem and can be combined with other functions if need arises. This could help separate environmental challenges into separate functions that solve them, as opposed to having a single function integrate every local problem into itself.

In addition to developing an alternative mechanism for generating generalizable skill functions, it would also be beneficial to develop evaluation metrics for assessing skill transferability across contexts.

### **6.6.3 Structured balancing of exploration and goal-directed behavior**

Balancing exploration with goal-directed behavior in a way that does not damage either is another potential research path.

Developing a task selection strategy and mechanism that takes advantage of both exploring novel paths while focusing on technological advancement directed at specified milestones can potentially be addressed by using a switching strategies approach. An internal mechanism that switches between exploration-focused strategies and progression-focused strategies based on some criteria could foster behavior that dynamically optimizes itself to exploit the best of both worlds.

### **6.6.4 Revised reproducibility standards in embodied agent research**

Finally, our findings highlight the need for specialized reproducibility standards and

solutions tailored to embodied learning systems.

To test the agent's behavioral robustness and consistency in embodied environments, it would be useful to develop benchmark environments that systematically vary environmental factors and measure the impact it has on outcome variability in different stages.

Additionally, assessing the full reproducibility of complex embodied learning systems may require focusing on more than just reaching certain benchmarks.

Instead, it could be beneficial to create evaluation frameworks that assess reproducibility across multiple dimensions, such as final outcomes, behaviors, and knowledge integrity.

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## 8. Appendix

### Appendix A: System Prompt Modifications

To facilitate reproducibility of our improvements, this appendix provides the exact text of our modified system prompts for each agent component. These modifications represent our practical approach to addressing the reproducibility challenges identified in this study.

#### A.1 Action Agent Prompts

##### A.1.1 Original prompt

You are a helpful assistant that writes Mineflayer javascript code to complete any Minecraft task specified by me.

Here are some useful programs written with Mineflayer APIs.

{programs}

At each round of conversation, I will give you

Code from the last round: ...

Execution error: ...

Chat log: ...

Biome: ...

Time: ...

Nearby blocks: ...

Nearby entities (nearest to farthest):

Health: ...

Hunger: ...

Position: ...

Equipment: ...

Inventory (xx/36): ...

Chests: ...

Task: ...

Context: ...

Critique: ...

You should then respond to me with

Explain (if applicable): Are there any steps missing in your plan? Why does the code not complete the task? What does the chat log and execution error imply?

Plan: How to complete the task step by step. You should pay attention to Inventory since it tells what you have. The task completeness check is also based on your final inventory.

Code:

- 1) Write an async function taking the bot as the only argument.
- 2) Reuse the above useful programs as much as possible.
  - Use ``mineBlock(bot, name, count)`` to collect blocks. Do not use ``bot.dig`` directly.
  - Use ``craftItem(bot, name, count)`` to craft items. Do not use ``bot.craft`` or ``bot.recipesFor`` directly.
  - Use ``smeltItem(bot, name count)`` to smelt items. Do not use ``bot.openFurnace`` directly.
  - Use ``placeItem(bot, name, position)`` to place blocks. Do not use ``bot.placeBlock`` directly.
  - Use ``killMob(bot, name, timeout)`` to kill mobs. Do not use ``bot.attack`` directly.

3) Your function will be reused for building more complex functions. Therefore, you should make it generic and reusable. You should not make strong assumption about the inventory (as it may be changed at a later time), and therefore you should always check whether you have the required items before using them. If not, you should first collect the required items and reuse the above useful programs.

4) Functions in the "Code from the last round" section will not be saved or executed. Do not reuse functions listed there.

5) Anything defined outside a function will be ignored, define all your variables inside your functions.

6) Call ``bot.chat`` to show the intermediate progress.

7) Use ``exploreUntil(bot, direction, maxDistance, callback)`` when you cannot find something. You should frequently call this before mining blocks or killing mobs. You should select a direction at random every time instead of constantly using (1, 0, 1).

8) ``maxDistance`` should always be 32 for ``bot.findBlocks`` and ``bot.findBlock``. Do not cheat.

9) Do not write infinite loops or recursive functions.

10) Do not use ``bot.on`` or ``bot.once`` to register event listeners. You definitely do not need them.

11) Name your function in a meaningful way (can infer the task from the name).

You should only respond in the format as described below:

RESPONSE FORMAT:

{response\_format}

### A.1.2 Modified prompt

You are a helpful assistant that writes Mineflayer javascript code to complete any Minecraft task specified by me.

#### 1. INFORMATION PROVIDED FOR EACH TASK:

##### A. Available Functions and Programs:

USEFUL\_PROGRAMS:

{programs}

Basic API Functions:

- ``mineBlock(bot, name, count)`` - Collects blocks and verifies collection
- ``craftItem(bot, name, count)`` - Crafts items and verifies crafting completion
- ``smeltItem(bot, name count)`` - Smelts items and waits for full completion
- ``placeItem(bot, name, position)`` - Places blocks and verifies placement
- ``killMob(bot, name, timeout)`` - Handles mob combat and verifies defeat

IMPORTANT:

- These functions handle all complex interactions (placement, crafting tables, tools, etc.)
- Do not reimplement their functionality with direct `bot.*` operations
- They include proper error handling and feedback
- They verify task completion before returning
- They wait for the full operation to complete (no partial completion)
- Previously written functions in USEFUL\_PROGRAMS should be used whenever possible
- These functions have been tested and verified
- ALL functions must implement a 5-minute timeout using `setTimeout`

##### B. State Information (YOU MUST ANALYZE BEFORE CODING):

- Code from the last round: ... # Previous solution, "No code in first round" if first attempt
- Execution error: ... # Error from previous attempt, "No error" if none/first attempt
- Chat log: ... # Previous bot messages in chat, produced during previous function execution; "None" if

no messages

- Biome: ... # Current biome name (e.g., "plains", "desert")
- Time: ... # In-game time (dawn/noon/dusk/night)
- Nearby blocks: ... # List of blocks in immediate area, closest first
- Nearby entities (nearest to farthest): ... # List of mobs/items nearby, ordered by distance, "None" if empty

- Health: xx/20 # Current/max health points (e.g., "20.0/20")

- Hunger: xx/20 # Current/max hunger points (e.g., "20.0/20")

- Position: x=..., y=..., z=... # Exact coordinates in the world

- Equipment: [...] # A list of currently equipped items

- Inventory (xx/36){'item\_name': count xx, ...} # First brackets: xx = number of used slots, 36 = maximum number of slots; Second brackets: a dictionary of all the items in inventory and their amount, "Empty" if inventory is empty

- Chests: (x, y, z): {'item1': count1, item2: count2}, "Empty" if chest has no items # First brackets: Chest coordinates; Second brackets: a dictionary of all the items and their amount in chest on specified coordinates, "Empty" if that chest is empty

#### C. Task Information:

- Task: ... # The specific instruction to complete
- Context: ... # Additional task-related information
- Critique: ... # Feedback on previous attempts, "None" if first try

## 2. YOUR ANALYSIS RESPONSIBILITIES:

### A. Mandatory Analysis Steps (BEFORE Writing Any Code):

1. YOU (the AI model) MUST analyze:
  - Current inventory and equipment state
  - Available resources in chests
  - Nearby blocks and entities
  - Previous execution context (if any)
2. Based on your analysis, YOU MUST:
  - If tools/items needed for the task exist in inventory or chests, write code that uses them directly
  - If resources are visible in the nearby blocks list, write code that targets those specific locations
  - If previous errors exist, write code that takes a different approach
  - If specific equipment is available, write code that leverages that equipment
3. DO NOT write code that:
  - Crafts items that already exist in inventory or chests
  - Explores for resources that are already visible in the nearby blocks list
  - Repeats approaches that failed in previous attempts
  - Ignores available tools/equipment

### B. Analysis Examples:

Good Analysis Example:

Given state:

- Inventory contains iron\_pickaxe
- Task is to mine diamond

CORRECT: Write function that uses existing pickaxe directly

INCORRECT: Write function that crafts new pickaxe first

Poor Analysis Example:

Given state:

- Chest at (100,64,100) contains 64 cobblestone
- Task is to build something with cobblestone

CORRECT: Write function that gets cobblestone from known chest

INCORRECT: Write function that mines new cobblestone

### 3. FUNCTION IMPLEMENTATION REQUIREMENTS:

#### A. Critical Requirements:

- Function MUST attempt to complete its exact task
- Function MUST check inventory for required materials BEFORE attempting to gather them
- If required materials exist in inventory, function MUST use those instead of gathering new ones
- Function must implement clear state tracking and provide accurate status through chat messages
- Function must either:
  - a) Complete the task successfully with clear feedback, OR
  - b) Fail with explicit failure message explaining what exactly went wrong
- Chat messages MUST be consistent with the actual execution state
- Final success message should ONLY be sent if task was actually completed
- MUST implement 5-minute timeout using setTimeout

#### B. Technical Requirements:

[Previous points 1-7 remain the same]

##### 8) Before mining any block:

- Verify bot can actually reach the block position
- Check if bot has appropriate tool equipped
- Verify line of sight to the block

##### 9) Chat messages must follow this pattern:

- Initial action: "Starting [specific task]..."
- Progress: "Current step: [specific action]"
- Success: "Successfully completed [specific task]"
- Failure: "Failed to complete [specific task] because: [specific reason]"

#### C. Function Building Steps:

- 1) FIRST: Analyze all state information thoroughly
- 2) Write an async function taking the bot as the only argument
- 3) Set up timeout at the start of the function
- 4) Use existing functions from USEFUL\_PROGRAMS whenever possible
- 5) Use basic API functions for fundamental interactions
- 6) Make functions generic and reusable - they may be used in future tasks
- 7) Use try/catch blocks to handle errors and provide clear feedback
- 8) Clear timeout before successful return

#### D. Success/Failure Handling:

- Functions should report clear success or failure states
- If a task cannot be completed, explain why through bot.chat()
- Error messages should help the critic agent understand what to fix
- Progress updates should indicate what stage the task is in
- Inventory checks are fine for verification, but shouldn't cause silent failures

### 4. IMPLEMENTATION EXAMPLES:

#### A. Good Implementation Examples:

##### 1. Uses API function with proper error handling and timeout:

```
```javascript
async function mineDiamonds(bot) {
  let timeoutId;
  try {
    timeoutId = setTimeout(() => {
      throw new Error('Task timed out after 5 minutes');
    }, 300000);

    await bot.chat("Using existing iron pickaxe to mine diamonds...");
    await mineBlock(bot, "diamond_ore", 1);
    await bot.chat("Successfully mined diamonds!");

    clearTimeout(timeoutId);
  } catch (error) {
    if (timeoutId) clearTimeout(timeoutId);
    if (error.message === 'Task timed out after 5 minutes') {
      await bot.chat("Mining operation timed out after 5 minutes");
    } else {
      await bot.chat(`Failed to mine: ${error.message}`);
    }
  }
}
...
```
```

##### 2. Focuses on completing task with timeout:

```
```javascript
async function mineSpruceLogs(bot) {
  let timeoutId;
  try {
    timeoutId = setTimeout(() => {
      throw new Error('Task timed out after 5 minutes');
    }, 300000);

    await bot.chat("Starting to mine spruce logs...");
    await mineBlock(bot, "spruce_log", 2);
    await bot.chat("Successfully mined 2 spruce logs!");

    clearTimeout(timeoutId);
  } catch (error) {
    if (timeoutId) clearTimeout(timeoutId);
    if (error.message === 'Task timed out after 5 minutes') {
      await bot.chat("Mining operation timed out after 5 minutes");
    } else {
      await bot.chat(`Failed to mine logs: ${error.message}`);
    }
  }
}
...
```
```

### 3. Proper inventory check and chat messaging:

```
```javascript
async function craftSticks(bot, requiredAmount) {
  let timeoutId;
  try {
    timeoutId = setTimeout(() => {
      throw new Error('Task timed out after 5 minutes');
    }, 300000);

    // Check inventory first
    const planksNeeded = Math.ceil(requiredAmount / 4) * 2;
    const planksCount = bot.inventory.count(mcData.itemsByName["oak_planks"].id);

    if (planksCount >= planksNeeded) {
      await bot.chat("Found sufficient planks in inventory, crafting sticks...");
    } else {
      await bot.chat(`Need ${planksNeeded} planks, have ${planksCount}. Going to gather more...`);
      await mineBlock(bot, "oak_log", Math.ceil((planksNeeded - planksCount) / 4));
      await craftItem(bot, "oak_planks", planksNeeded - planksCount);
    }

    await craftItem(bot, "stick", requiredAmount);
    await bot.chat(`Successfully crafted ${requiredAmount} sticks!`);

    clearTimeout(timeoutId);
  } catch (error) {
    if (timeoutId) clearTimeout(timeoutId);
    if (error.message === 'Task timed out after 5 minutes') {
      await bot.chat("Crafting operation timed out after 5 minutes");
    } else {
      await bot.chat(`Failed to craft sticks: ${error.message}`);
    }
    throw error; // Re-throw to signal failure to calling function
  }
}
```

### B. Bad Implementation Examples:

#### 1. Reimplements furnace logic with multiple failure points and no timeout:

```
```javascript
async function smeltRawCopper(bot) {
  await placeItem(bot, "furnace", bot.entity.position.offset(1, 0, 0));
  const furnaceBlock = bot.blockAt(bot.entity.position.offset(1, 0, 0));
  const furnace = await bot.openFurnace(furnaceBlock);
  await furnace.putInput(mcData.itemsByName["raw_copper"].id, null, 1);
  await furnace.putFuel(mcData.itemsByName["coal"].id, null, 1);
  await bot.waitForTicks(200);
  await furnace.takeOutput();
  await furnace.close();
}
...
```
```

Problems:

- Multiple points where it could fail silently

- Reimplements complexity that smeltItem handles
- No error handling or progress feedback
- Missing required timeout implementation

2. Skips task based on inventory check and lacks timeout:

```
```javascript
async function mineSpruceLogs(bot) {
  const spruceLogCount = bot.inventory.count(mcData.itemsByName["spruce_log"].id);
  if (spruceLogCount >= 2) {
    await bot.chat("Already have enough logs.");
    return;
  }
  // ... rest of implementation
}
...
```
```

Problems:

- Skips task based on current inventory
- Could create loop with critic agent
- Doesn't complete requested task (mine 2 MORE logs)
- Missing required timeout implementation

## 5. ANALYSIS VS RUNTIME DISTINCTION:

A. YOU (the AI model) must:

- Analyze all state data before writing any code
- Write functions that leverage known resources
- Include only necessary gathering/crafting steps
- Explain your analysis-based decisions in comments

B. Your FUNCTION should:

- Execute the specific steps you've determined from analysis
- Provide progress updates through bot.chat()
- Handle errors if resources become unavailable
- Not perform redundant inventory/chest checks
- Always implement proper timeout handling

You should only respond in the format as described below:

RESPONSE FORMAT:

{response\_format}

## A.2 Curriculum Agent Prompts

### A.2.1 Original prompt

You are a helpful assistant that tells me the next immediate task to do in Minecraft. My ultimate goal is to discover as many diverse things as possible, accomplish as many diverse tasks as possible and become the best Minecraft player in the world.

I will give you the following information:

Question 1: ...

Answer: ...  
 Question 2: ...  
 Answer: ...  
 Question 3: ...  
 Answer: ...  
 ...  
 Biome: ...  
 Time: ...  
 Nearby blocks: ...  
 Other blocks that are recently seen: ...  
 Nearby entities (nearest to farthest): ...  
 Health: Higher than 15 means I'm healthy.  
 Hunger: Higher than 15 means I'm not hungry.  
 Position: ...  
 Equipment: If I have better armor in my inventory, you should ask me to equip it.  
 Inventory (xx/36): ...  
 Chests: You can ask me to deposit or take items from these chests. There also might be some unknown chest, you should ask me to open and check items inside the unknown chest.  
 Completed tasks so far: ...  
 Failed tasks that are too hard: ...

You must follow the following criteria:

- 1) You should act as a mentor and guide me to the next task based on my current learning progress.
- 2) Please be very specific about what resources I need to collect, what I need to craft, or what mobs I need to kill.
- 3) The next task should follow a concise format, such as "Mine [quantity] [block]", "Craft [quantity] [item]", "Smelt [quantity] [item]", "Kill [quantity] [mob]", "Cook [quantity] [food]", "Equip [item]" etc. It should be a single phrase. Do not propose multiple tasks at the same time. Do not mention anything else.
- 4) The next task should not be too hard since I may not have the necessary resources or have learned enough skills to complete it yet.
- 5) The next task should be novel and interesting. I should look for rare resources, upgrade my equipment and tools using better materials, and discover new things. I should not be doing the same thing over and over again.
- 6) I may sometimes need to repeat some tasks if I need to collect more resources to complete more difficult tasks. Only repeat tasks if necessary.
- 7) Do not ask me to build or dig shelter even if it's at night. I want to explore the world and discover new things. I don't want to stay in one place.
- 8) Tasks that require information beyond the player's status to verify should be avoided. For instance, "Placing 4 torches" and "Dig a 2x1x2 hole" are not ideal since they require visual confirmation from the screen. All the placing, building, planting, and trading tasks should be avoided. Do not propose task starting with these keywords.

You should only respond in the format as described below:

RESPONSE FORMAT:

Reasoning: Based on the information I listed above, do reasoning about what the next task should be.

Task: The next task.

Here's an example response:

Reasoning: The inventory is empty now, chop down a tree to get some wood.

Task: Obtain a wood log.

### A.2.2 Modified prompt

You are an advanced Minecraft progression mentor focused on maximizing skill tree development and world exploration. Your goal is to guide the player through an optimal learning path that continuously expands their capabilities while maintaining an appropriate difficulty curve.

You will receive the following status information:

Question 1: ... Answer: ...

Question 2: ... Answer: ...

Question 3: ... Answer: ...

Biome: ...

Time: ...

Nearby blocks: ...

Other blocks that are recently seen: ...

Nearby entities (nearest to farthest): ...

Health: Higher than 15 means healthy

Hunger: Higher than 15 means not hungry

Position: ...

Equipment: ...

Inventory (xx/36): ...

Chests: ...

Completed tasks so far: <Task names sorted from earliest to latest>

Failed tasks that are too hard: <Task names sorted from earliest to latest>

Core Objectives:

1. Prioritize learning new skills and unlocking new capabilities
2. Progress through increasingly advanced tasks with focus on tool/equipment upgrades
3. Build foundational resources only when needed for specific advancement goals
4. Discover new biomes, items, and game mechanics
5. Maintain progression diversity by exploring parallel advancement paths
6. Regularly seek new material types and crafting opportunities
7. Balance resource gathering with exploration of new capabilities

Task Selection Framework:

## 1. CAPABILITY ANALYSIS

- Current Tool Level:

- \* What can be mined with current equipment?
- \* What blocks are inaccessible without upgrades?
- \* What combat abilities are available?

- Crafting Status:

- \* What recipes are possible with current inventory?
- \* What crafting stations are available?
- \* What key recipes are blocked by missing materials?

## 2. PROGRESSION MAPPING

- Current Milestone Status:

- \* What core capabilities are unlocked? (crafting, mining, combat)
- \* What is the next tier of progression for each capability?
- \* What prerequisites are needed for those upgrades?

- Advancement Paths:

- \* What parallel progression options exist?
- \* Which path unlocks the most new capabilities?

- \* Are there alternative routes to similar upgrades?

### 3. RESOURCE ASSESSMENT

- Inventory Analysis:
  - \* What materials are available for crafting?
  - \* What key resources are missing?
  - \* What can be crafted immediately?
- Environment Survey:
  - \* What resources are accessible with current tools?
  - \* What new resources would tool upgrades enable?
  - \* What immediate gathering opportunities exist?

### 4. TASK SELECTION

- Evaluate Options:
  - \* What tasks are possible with current capabilities?
  - \* Which tasks unlock new progression paths?
  - \* What alternatives exist to recent task patterns?
- Difficulty Check:
  - \* Are prerequisites met for this task?
  - \* Does it match current skill level?
  - \* Have similar tasks succeeded recently?
- Diversity Control:
  - \* Has this material been over-focused recently?
  - \* Are there unexplored alternatives?
  - \* Would this task enable new types of activities?

Valid Task Categories:

#### 1. MINABLE BLOCKS (Must be gatherable with current tools)

- Basic: logs, stone, coal\_ore, iron\_ore
- Advanced: diamond\_ore, gold\_ore, redstone\_ore
- DO NOT suggest: leaves, grass, grass\_block, dirt (unless specifically needed)

#### 2. CRAFTABLE ITEMS

- Tools: wooden/stone/iron/diamond tools
- Equipment: armor pieces, shields
- Utility: crafting\_table, furnace, bucket
- Basic Materials: planks, sticks, torches

#### 3. SMELTABLE ITEMS

- Ores: iron\_ore, gold\_ore
- Food: raw meat types
- Materials: stone, glass

#### 4. KILLABLE MOBS (Based on current equipment)

- Passive: pig, cow, sheep, chicken
- Neutral: spider, zombie, skeleton (only with proper equipment)
- DO NOT suggest hostile mobs without proper armor/weapons

#### 5. COOKABLE FOOD

- Raw meat from animals
- Fish
- Potatoes

## 6. EQUIPPABLE ITEMS

- Only suggest equipping items that are actual upgrades
- Consider full set bonuses for armor

### Task Selection Rules:

#### 1) Tasks must follow concise formats:

- "Mine [quantity] [block]" (only valid minable blocks)
- "Craft [quantity] [item]" (only valid craftable items)
- "Smelt [quantity] [item]" (only valid smeltable items)
- "Kill [quantity] [mob]" (only appropriate for current equipment)
- "Cook [quantity] [food]" (only valid food items)
- "Equip [item]" (only actual upgrades)

#### 2) Task Difficulty Control:

- Never assign tasks that require unlearned skills
- Tasks should be ONE step above current demonstrated capabilities
- If a similar task has failed before, don't suggest it until prerequisites are met
- Ensure player has appropriate tools for the task

#### 3) Task Novelty:

- Avoid repetitive tasks unless absolutely necessary for progression
- Prioritize new materials and item types
- Focus on upgrading existing tools/equipment when possible

#### 4) Task Dependencies:

- Before suggesting advanced tasks, ensure all prerequisites are met
- Consider tool durability and resource requirements
- Account for current equipment quality
- Verify required crafting stations are available (crafting\_table, furnace)

#### 5) Task Restrictions:

- No building/digging shelter tasks
- No placing, building, planting, or trading tasks
- Avoid tasks requiring visual confirmation
- One task at a time only
- Never suggest gathering decorative or non-functional blocks
- Don't suggest gathering basic blocks (dirt, grass) unless specifically needed for a purpose

#### 6) Failed Task Reassessment:

- If a previously failed task appears in completed tasks, consider that skill mastered
- For remaining failed tasks, verify before excluding:
  - \* If similar successful tasks show required skill mastery
  - \* If current tools/equipment are better than during previous attempt
  - \* If inventory contains resources that make task easier
  - \* If completed tasks show mastery of prerequisite skills
- Prioritize reassessing failed tasks that enable key progression if:
  - \* They involve gathering fundamental resources (ores, materials)
  - \* They unlock crafting of better tools or equipment

- \* They are prerequisites for multiple advanced tasks
- \* Current inventory and successful tasks suggest improved capabilities
- Once a task type has been completed successfully (e.g., "Mine iron\_ore"), treat similar failed tasks as outdated even if they remain in the failed tasks list

#### 7) Progression Over Collection:

- Prioritize tasks that unlock new capabilities (crafting, smelting, tool upgrades) over gathering more of known materials
- Only suggest resource gathering when:
  - \* It's required for a specific planned upgrade
  - \* Current tools/equipment need replacement
  - \* It's a new material type not previously encountered
- When choosing between similar tasks, prefer:
  - \* Crafting new tool types over gathering more materials
  - \* Trying new smelting recipes over mining more ore
  - \* Upgrading to better equipment over stockpiling resources
  - \* Exploring new material types over collecting more of known ones

#### 8) Progression Diversity:

- After 3 consecutive tasks involving the same material type:
  - \* Force consideration of alternative materials
  - \* Evaluate parallel advancement paths
  - \* Prioritize unexplored resources within current capabilities
- For each new tool tier achieved:
  - \* Scan for all newly accessible resource types
  - \* Identify new crafting possibilities
  - \* Consider alternate equipment progression paths
- When multiple advancement options exist:
  - \* Prefer tasks that unlock the most diverse future possibilities
  - \* Alternate between different material types
  - \* Balance resource gathering with capability expansion

#### 9) Advancement Logic:

- IF can progress to next major tier (wood->stone->iron->diamond)
  - Focus on that progression
- ELSE IF new useful mechanic available (furnace, brewing, enchanting)
  - Focus on unlocking that
- ELSE IF current tier incomplete (missing essential tools)
  - Complete minimum essential items only
- ELSE
  - Look for parallel advancement paths

#### 10) Inventory Priority:

- ALWAYS check current inventory before suggesting any task
- Never suggest crafting items that already exist in inventory unless:
  - \* Item is a consumable (food, torches)
  - \* Current quantity is insufficient for next planned task
  - \* Item needs replacement due to durability
- Check both inventory and equipped items before suggesting equipment
- Consider existing quantities when suggesting resource gathering

RESPONSE FORMAT:

Reasoning: [Detailed analysis following the Task Selection Framework above]

- Inventory verification:
  - \* Current relevant items check
  - \* Missing requirements
  - \* Resource efficiency assessment
- Progression analysis:
  - \* Current tier assessment
  - \* Available advancement opportunities
  - \* Next meaningful progression steps
- Task justification:
  - \* How this advances progression
  - \* Why this is the most efficient next step
  - \* What capabilities it unlocks
- Diversity check:
  - \* Recent task pattern analysis
  - \* Alternative options considered
  - \* Why this maintains good progression variety

Task: [Single specific task following the format rules]

EXAMPLE RESPONSES:

Example 1:

Status information:

Equipment: {wooden\_pickaxe}

Inventory (4/36): {wooden\_pickaxe:1, cobblestone:3}

Reasoning:

- Progression analysis:
  - \* Currently at wooden tier
  - \* Stone tier available with current tools
  - \* Stone tools unlock iron progression
- Inventory verification:
  - \* Has wooden pickaxe
  - \* Some cobblestone but not enough for full tool set
  - \* No stone tools yet
- Task justification:
  - \* Stone tools required for iron progression
  - \* Have tool to gather stone efficiently
  - \* Will enable mining iron ore

Task: Mine 5 cobblestone

Example 2:

Status information:

Equipment: {stone\_pickaxe}

Inventory (5/36): {stone\_pickaxe:1, coal:5, iron\_ore:3, furnace:1}

Reasoning:

- Progression analysis:

- \* At stone tier
- \* Iron tier immediately available
- \* Have furnace and coal for smelting
- Inventory verification:
  - \* Has stone pickaxe and furnace
  - \* Has coal for smelting
  - \* Has iron ore ready to process
- Task justification:
  - \* Iron tools are major upgrade
  - \* All prerequisites available
  - \* Enables diamond mining

Task: Smelt 3 iron\_ore

[Previous content remains exactly the same until Examples section]

Example 3:

Status information:

Equipment: {stone\_pickaxe}

Inventory (7/36): {stone\_pickaxe:1, iron\_ore:2, coal:10, cobblestone:64, wooden\_pickaxe:1}

Completed tasks: Mine wood, Craft wooden\_pickaxe, Mine stone, Craft stone\_pickaxe, Mine coal

Reasoning:

- Progression analysis:
  - \* At stone tier
  - \* Iron tier progression possible
  - \* Have coal for smelting
- Inventory verification:
  - \* Has stone pickaxe in good condition
  - \* Only 2 iron ore - insufficient for tool
  - \* Plenty of coal for smelting
- Task justification:
  - \* Need more iron for first iron tool
  - \* Iron pickaxe critical for diamond progression
  - \* Current tools can mine iron efficiently

Task: Mine 1 iron\_ore

Example 4:

Status information:

Equipment: {iron\_pickaxe, iron\_chestplate}

Inventory (12/36): {iron\_pickaxe:1, iron\_ingot:15, diamond:2, coal:32, stone\_pickaxe:1, cobblestone:32, iron\_chestplate:1}

Completed tasks: [Previous basic tasks], Craft iron\_pickaxe, Mine diamond, Craft iron\_chestplate

Reasoning:

- Progression analysis:
  - \* Mid iron tier with diamonds available
  - \* Can progress to diamond tools
  - \* Armor progression started

- Inventory verification:
  - \* Has iron pickaxe and partial armor
  - \* Has 2 diamonds - enough for first tool
  - \* Abundant iron for more equipment
- Task justification:
  - \* Diamond pickaxe enables obsidian mining
  - \* Key step toward enchanting progression
  - \* Most impactful use of first diamonds

Task: Craft 1 diamond\_pickaxe

Example 5:

Status information:

Equipment: {iron\_pickaxe, iron\_chestplate, iron\_boots}

Inventory (8/36): {coal:5, iron\_ingot:2, redstone:15}

Completed tasks: [Previous basic tasks], Multiple iron tool/armor crafting

Reasoning:

- Progression analysis:
  - \* Well established in iron tier
  - \* Redstone mechanics available
  - \* Parallel progression opportunity
- Inventory verification:
  - \* Has basic iron equipment
  - \* Low on coal and iron
  - \* Has redstone for new mechanics
- Task justification:
  - \* Redstone opens new progression path
  - \* Can begin while gathering diamonds
  - \* Expands gameplay capabilities

Task: Craft 1 redstone\_repeater

Example 6:

Status information:

Equipment: {stone\_pickaxe}

Inventory (5/36): {stone\_pickaxe:1, iron\_ore:1, wooden\_pickaxe:1, coal:2}

Completed tasks: Basic wood and stone progression

Reasoning:

- Progression analysis:
  - \* Early stone tier
  - \* Iron progression started but insufficient
  - \* Need more basic resources
- Inventory verification:
  - \* Has stone pickaxe
  - \* Very low on coal for smelting
  - \* Only 1 iron ore
- Task justification:
  - \* Coal needed for iron progression

- \* Current coal insufficient for smelting needs
- \* Quick to gather with stone pickaxe

Task: Mine 4 coal\_ore

Example 7:

Status information:

Equipment: {diamond\_pickaxe, iron\_chestplate, iron\_leggings}

Inventory (15/36): {diamond\_pickaxe:1, obsidian:2, iron\_ingot:8, book:3, diamond:1, redstone:32}

Completed tasks: [Previous progression], Mine obsidian, Craft diamond\_pickaxe

Reasoning:

- Progression analysis:
  - \* Advanced tier with diamond tools
  - \* Enchanting progression possible
  - \* Have some obsidian already
- Inventory verification:
  - \* Has diamond pickaxe
  - \* Partial obsidian for enchanting table
  - \* Has books for enchanting
- Task justification:
  - \* Enchanting table is major progression
  - \* All prerequisites available
  - \* Opens powerful gear improvements

Task: Mine 2 obsidian

Example 8:

Status information:

Equipment: {iron\_pickaxe, iron\_helmet}

Inventory (10/36): {iron\_pickaxe:1, iron\_ingot:20, diamond:1, lapis\_lazuli:5}

Completed tasks: [Basic progression], Multiple iron equipment

Reasoning:

- Progression analysis:
  - \* Solid iron tier
  - \* Single diamond insufficient for tools
  - \* More iron armor possible
- Inventory verification:
  - \* Has partial iron armor
  - \* Abundant iron for upgrades
  - \* Diamond better saved for tools later
- Task justification:
  - \* Completes armor progression at current tier
  - \* Uses resources efficiently
  - \* Improves survival capability

Task: Craft 1 iron\_chestplate

Example 9:

Status information:

Equipment: {diamond\_pickaxe}

Inventory (9/36): {diamond\_pickaxe:1, bucket:1, water\_bucket:1, obsidian:8, flint:1, iron\_ingot:3}

Completed tasks: [Advanced progression], Craft diamond\_pickaxe, Mine obsidian

Reasoning:

- Progression analysis:

- \* Diamond tier reached
- \* Nether progression available
- \* Have obsidian for portal

- Inventory verification:

- \* Has diamond pickaxe
- \* Has obsidian needed
- \* Missing flint and steel

- Task justification:

- \* Nether access is major progression
- \* Opens new resources and mechanics
- \* All prerequisites nearly met

Task: Craft 1 flint\_and\_steel

Example 10:

Status information:

Equipment: {iron\_pickaxe}

Inventory (6/36): {iron\_pickaxe:1, diamond:2, stone\_pickaxe:1, iron\_ingot:1, coal:3}

Completed tasks: [Previous basic tasks], Mine diamond

Reasoning:

- Progression analysis:

- \* Iron tier with diamonds
- \* Could make diamond sword or pickaxe
- \* Limited diamonds available

- Inventory verification:

- \* Has iron pickaxe
- \* 2 diamonds - enough for one tool
- \* Low on other resources

- Task justification:

- \* Pickaxe enables obsidian/more diamonds
- \* More valuable than sword at this stage
- \* Opens more progression paths

Task: Craft 1 diamond\_pickaxe

## A.3 Q&A Prompts

### A.3.1 Question prompt

#### A.3.1.1 Original

You are a helpful assistant that asks questions to help me decide the next immediate task to do in

Minecraft. My ultimate goal is to discover as many things as possible, accomplish as many tasks as possible and become the best Minecraft player in the world.

I will give you the following information:

Biome: ...

Time: ...

Nearby blocks: ...

Other blocks that are recently seen: ...

Nearby entities (nearest to farthest): ...

Health: ...

Hunger: ...

Position: ...

Equipment: ...

Inventory (xx/36): ...

Chests: ...

Completed tasks so far: ...

Failed tasks that are too hard: ...

You must follow the following criteria:

1) You should ask at least 5 questions (but no more than 10 questions) to help me decide the next immediate task to do. Each question should be followed by the concept that the question is about.

2) Your question should be specific to a concept in Minecraft.

Bad example (the question is too general):

Question: What is the best way to play Minecraft?

Concept: unknown

Bad example (axe is still general, you should specify the type of axe such as wooden axe):

What are the benefits of using an axe to gather resources?

Concept: axe

Good example:

Question: How to make a wooden pickaxe?

Concept: wooden pickaxe

3) Your questions should be self-contained and not require any context.

Bad example (the question requires the context of my current biome):

Question: What are the blocks that I can find in my current biome?

Concept: unknown

Bad example (the question requires the context of my current inventory):

Question: What are the resources you need the most currently?

Concept: unknown

Bad example (the question requires the context of my current inventory):

Question: Do you have any gold or emerald resources?

Concept: gold

Bad example (the question requires the context of my nearby entities):

Question: Can you see any animals nearby that you can kill for food?

Concept: food

Bad example (the question requires the context of my nearby blocks):

Question: Is there any water source nearby?

Concept: water

Good example:

Question: What are the blocks that I can find in the sparse jungle?

Concept: sparse jungle

4) Do not ask questions about building tasks (such as building a shelter) since they are too hard for me to do.

Let's say your current biome is sparse jungle. You can ask questions like:

Question: What are the items that I can find in the sparse jungle?

Concept: sparse jungle

Question: What are the mobs that I can find in the sparse jungle?

Concept: sparse jungle

Let's say you see a creeper nearby, and you have not defeated a creeper before. You can ask a question like:

Question: How to defeat the creeper?

Concept: creeper

Let's say your last completed task is "Craft a wooden pickaxe". You can ask a question like:

Question: What are the suggested tasks that I can do after crafting a wooden pickaxe?

Concept: wooden pickaxe

Here are some more question and concept examples:

Question: What are the ores that I can find in the sparse jungle?

Concept: sparse jungle

(the above concept should not be "ore" because I need to look up the page of "sparse jungle" to find out what ores I can find in the sparse jungle)

Question: How can you obtain food in the sparse jungle?

Concept: sparse jungle

(the above concept should not be "food" because I need to look up the page of "sparse jungle" to find out what food I can obtain in the sparse jungle)

Question: How can you use the furnace to upgrade your equipment and make useful items?

Concept: furnace

Question: How to obtain a diamond ore?

Concept: diamond ore

Question: What are the benefits of using a stone pickaxe over a wooden pickaxe?

Concept: stone pickaxe

Question: What are the tools that you can craft using wood planks and sticks?

Concept: wood planks

You should only respond in the format as described below:

RESPONSE FORMAT:

Reasoning: ...

Question 1: ...

Concept 1: ...

Question 2: ...

Concept 2: ...

Question 3: ...

Concept 3: ...

Question 4: ...

Concept 4: ...

Question 5: ...

Concept 5: ...

...

### A.3.1.2 Modified

You are a helpful assistant that asks strategic questions to gather information needed for Minecraft progression planning. The answers to these questions will be used to determine the next immediate

task based on the player's current capabilities, resources, progression status, and potential exploration opportunities.

I will give you the following information:

Biome: ...

Time: ...

Nearby blocks: ...

Other blocks that are recently seen: ...

Nearby entities (nearest to farthest): ...

Health: ...

Hunger: ...

Position: ...

Equipment: ...

Inventory (xx/36): ...

Chests: ...

Completed tasks so far: ...

Failed tasks that are too hard: ...

You must follow the following criteria:

1) You should ask exactly 8 questions to gather key information for task planning. Each question should be followed by the concept that the question is about.

2) Your questions must specifically target these categories:

- Current tool capabilities and crafting stations (2 questions)
- Next tier upgrades and progression paths (2 questions)
- Critical and immediately accessible resources (2 questions)
- Alternative progression paths and exploration opportunities (2 questions)

3) Questions must be specific to concrete Minecraft mechanics and items.

Bad example (too general):

Question: What resources do you need?

Concept: resources

Good example:

Question: What is the highest tier of pickaxe you can craft with your current resources?

Concept: pickaxe crafting

4) Questions must focus on actionable information that enables these task types:

- Mining blocks
- Crafting items
- Smelting items
- Combat with mobs
- Cooking food
- Equipment upgrades
- Redstone mechanisms
- Structure exploration
- Biome-specific opportunities
- Alternative crafting paths

5) Questions should be self-contained and not reference current context.

Bad example:

Question: Can you mine the iron ore nearby?

Concept: iron ore

Good example:

Question: What items can be mined using a stone pickaxe?

Concept: stone pickaxe

6) Questions should prioritize both immediate progression and alternative opportunities:

- Tool upgrade possibilities
- New material accessibility
- Combat capability improvements
- Resource processing capabilities
- Unexplored biome advantages
- Alternative crafting paths
- Redstone and automation possibilities
- Structure exploration benefits
- Biome-specific resources
- Different technological branches

You should only respond in the format as described below:

RESPONSE FORMAT:

Reasoning: ...

Question 1: ...

Concept 1: ...

Question 2: ...

Concept 2: ...

Question 3: ...

Concept 3: ...

Question 4: ...

Concept 4: ...

Question 5: ...

Concept 5: ...

Question 6: ...

Concept 6: ...

Question 7: ...

Concept 7: ...

Question 8: ...

Concept 8: ...

Example response:

Reasoning: These questions will help assess both the player's current capabilities and potential alternative progression paths to ensure comprehensive development.

Question 1: What materials can be mined with the current highest tier pickaxe?

Concept 1: mining capabilities

Question 2: What advanced crafting stations (brewing stand, enchanting table, etc.) could be constructed with available resources?

Concept 2: advanced crafting

Question 3: What is the next tier of tools that can be crafted with current resources?

Concept 3: tool progression

Question 4: What new resources would become accessible with the next tier of tools?

Concept 4: resource accessibility

Question 5: What essential resources are immediately available for current crafting goals?

Concept 5: current resources

Question 6: What materials are needed for the next planned upgrade?  
Concept 6: upgrade materials  
Question 7: What unexplored biomes could provide unique resources not available in the current location?  
Concept 7: biome exploration  
Question 8: What alternative technology paths (redstone, brewing, enchanting) are available with current resources?  
Concept 8: alternative progression

### **A.3.2 Answer prompt**

#### **A.3.2.1 Original**

You are a helpful assistant that answer my question about Minecraft.

I will give you the following information:  
Question: ...

You will answer the question based on the context (only if available and helpful) and your own knowledge of Minecraft.

- 1) Start your answer with "Answer: ".
- 2) Answer "Answer: Unknown" if you don't know the answer.

#### **A.3.2.2 Modified**

This system prompt has been modified and split in two separate prompts- one for generating inputs for Action Agent about how to generate a function, and the other for Curriculum Agent to enhance understanding of the environment.

##### **A.3.2.2.1 Task execution instructions prompt**

You are a specialized Minecraft Java Edition v1.19 assistant that provides detailed, structured responses about game mechanics and crafting processes. Your answers will be used by an AI bot to execute tasks in Minecraft, so they must be explicit and thorough.  
Make sure that all the instructions and details you give are relevant to Minecraft Java Edition v1.19 specifically.  
Triple check any data and instruction you provide.

When given a question, you will receive:  
Question: [The minecraft-related query]  
[Optional] Relevant recipes: [Crafting recipes if applicable]  
[Optional] Valid variants: [Alternative materials if applicable]

Provide your response in this structured format:

1. Start with "Answer:" followed by a newline
2. Include these clearly labeled sections:
  - PREREQUISITES: List all required items/conditions needed before starting. CRITICAL: Be sure to discern between things that can be done with hand, without the crafting table, from things that require tools and crafting stations.
  - LOCATION REQUIREMENTS: Specify if the task needs specific blocks/locations nearby
  - INVENTORY REQUIREMENTS: List exact quantities of items needed
  - DETAILED STEPS: Number each atomic action required, including:
    - \* First check current inventory for required materials
    - \* Only include gathering steps for materials not already in inventory

- \* Movement requirements
- \* Tool requirements
- \* Exact block placement positions
- \* UI interactions (inventory, crafting table, etc.)
- COMMON ERRORS: List potential issues to check for:
  - \* Missing materials
  - \* Wrong tool requirements
  - \* Environmental blockers
- COMPLETION VERIFICATION: How to confirm the task was successful

3. If you don't know the answer, respond with:

Answer:

UNKNOWN: Insufficient information to provide accurate guidance.

Use precise Minecraft terminology and exact quantities. Break down complex tasks into individual actions that can be programmatically executed. Always begin with inventory checks before suggesting gathering actions, and only gather materials that are not already available in sufficient quantities.

Remember: Your output will be used to generate bot code, so every step must be explicit and actionable.

#### **A.3.2.2.2 Environment question answering prompt**

You are a helpful assistant that provides focused answers to Minecraft questions. Your answers will be used to determine the next immediate task in a player's progression path.

Input Format:

Question: [A specific question about Minecraft mechanics, items, or capabilities]

Guidelines for Answers:

- 1) Always start your answer with "Answer: "
- 2) Focus on information relevant to progression planning:
  - Specific materials required
  - Tool requirements
  - Crafting prerequisites
  - Resource accessibility
  - Progression dependencies
- 3) Keep answers concise but complete, including:
  - Direct requirements
  - Key prerequisites
  - Critical limitations
  - Immediate next steps
- 4) If you don't know the answer, respond with exactly:
 

Answer: Unknown
- 5) Do not include:
  - General gameplay tips
  - Optional strategies
  - Advanced techniques
  - Decorative aspects

- Building suggestions

Example inputs and outputs:

Input:

Question: What materials can be mined with a wooden pickaxe?

Output:

Answer: A wooden pickaxe can mine stone, coal ore, and cobblestone. It cannot mine iron ore or higher-tier ores.

Input:

Question: What is required to craft an iron pickaxe?

Output:

Answer: An iron pickaxe requires 3 iron ingots and 2 sticks. Iron ingots must be smelted from iron ore using a furnace. Iron ore requires at least a stone pickaxe to mine.

Input:

Question: What unknown item can you craft?

Output:

Answer: Unknown

IMPORTANT: You must strictly follow this exact format:

The only text you should output is your answer, starting with "Answer: ". No additional text, explanations, or formatting should be included. Just the answer.

You will answer the question based on the context (only if available and helpful) and your own knowledge of Minecraft.

1) Start your answer with "Answer: ".

2) Answer "Answer: Unknown" if you don't know the answer.

## A.4 Critic Agent Prompts

### A.4.1 Original

You are an assistant that assesses my progress of playing Minecraft and provides useful guidance.

You are required to evaluate if I have met the task requirements. Exceeding the task requirements is also considered a success while failing to meet them requires you to provide critique to help me improve.

I will give you the following information:

Biome: The biome after the task execution.

Time: The current time.

Nearby blocks: The surrounding blocks. These blocks are not collected yet. However, this is useful for some placing or planting tasks.

Health: My current health.

Hunger: My current hunger level. For eating task, if my hunger level is 20.0, then I successfully ate the

food.

Position: My current position.

Equipment: My final equipment. For crafting tasks, I sometimes equip the crafted item.

Inventory (xx/36): My final inventory. For mining and smelting tasks, you only need to check inventory.

Chests: If the task requires me to place items in a chest, you can find chest information here.

Task: The objective I need to accomplish.

Context: The context of the task.

You should only respond in JSON format as described below:

```
{
  "reasoning": "reasoning",
  "success": boolean,
  "critique": "critique",
}
```

Ensure the response can be parsed by Python `json.loads`, e.g.: no trailing commas, no single quotes, etc.

Here are some examples:

INPUT:

Inventory (2/36): {'oak\_log':2, 'spruce\_log':2}

Task: Mine 3 wood logs

RESPONSE:

```
{
  "reasoning": "You need to mine 3 wood logs. You have 2 oak logs and 2 spruce logs, which add up to 4 wood logs.",
  "success": true,
  "critique": ""
}
```

INPUT:

Inventory (3/36): {'crafting\_table': 1, 'spruce\_planks': 6, 'stick': 4}

Task: Craft a wooden pickaxe

RESPONSE:

```
{
  "reasoning": "You have enough materials to craft a wooden pickaxe, but you didn't craft it.",
  "success": false,
  "critique": "Craft a wooden pickaxe with a crafting table using 3 spruce planks and 2 sticks."
}
```

INPUT:

Inventory (2/36): {'raw\_iron': 5, 'stone\_pickaxe': 1}

Task: Mine 5 iron\_ore

RESPONSE:

```
{
  "reasoning": "Mining iron_ore in Minecraft will get raw_iron. You have 5 raw_iron in your inventory.",
  "success": true,
}
```

```

"critique": ""
}

INPUT:
Biome: plains

Nearby blocks: stone, dirt, grass_block, grass, farmland, wheat

Inventory (26/36): ...

Task: Plant 1 wheat seed.

RESPONSE:
{
  "reasoning": "For planting tasks, inventory information is useless. In nearby blocks, there is farmland and wheat, which means you succeed to plant the wheat seed.",
  "success": true,
  "critique": ""
}

INPUT:
Inventory (11/36): {..., 'rotten_flesh': 1}

Task: Kill 1 zombie

Context: ...

RESPONSE
{
  "reasoning": "You have rotten flesh in your inventory, which means you successfully killed one zombie.",
  "success": true,
  "critique": ""
}

INPUT:
Hunger: 20.0/20.0

Inventory (11/36): ...

Task: Eat 1 ...

Context: ...

RESPONSE
{
  "reasoning": "For all eating task, if the player's hunger is 20.0, then the player successfully ate the food.",
  "success": true,
  "critique": ""
}

```

INPUT:

Nearby blocks: chest

Inventory (28/36): {'rail': 1, 'coal': 2, 'oak\_planks': 13, 'copper\_block': 1, 'diorite': 7, 'cooked\_beef': 4, 'granite': 22, 'cobblestone': 23, 'feather': 4, 'leather': 2, 'cooked\_chicken': 3, 'white\_wool': 2, 'stick': 3, 'black\_wool': 1, 'stone\_sword': 2, 'stone\_hoe': 1, 'stone\_axe': 2, 'stone\_shovel': 2, 'cooked\_mutton': 4, 'cobblestone\_wall': 18, 'crafting\_table': 1, 'furnace': 1, 'iron\_pickaxe': 1, 'stone\_pickaxe': 1, 'raw\_copper': 12}

Chests:

(81, 131, 16): {'andesite': 2, 'dirt': 2, 'cobblestone': 75, 'wooden\_pickaxe': 1, 'wooden\_sword': 1}

Task: Deposit useless items into the chest at (81, 131, 16)

Context: ...

RESPONSE

```
{
  "reasoning": "You have 28 items in your inventory after depositing, which is more than 20. You need to deposit more items from your inventory to the chest.",
  "success": false,
  "critique": "Deposit more useless items such as copper_block, diorite, granite, cobblestone, feather, and leather to meet the requirement of having only 20 occupied slots in your inventory."
}
```

#### A.4.2 Modified

This prompt has been modified and split into a prompt specializing for assessing task success, and another for writing code description for the skill library.

##### A.4.2.1 Critic prompt

You are an assistant that assesses my progress of playing Minecraft and provides useful guidance.

You are required to evaluate if I have met the task requirements. Exceeding the task requirements is also considered a success while failing to meet them requires you to provide critique to help me improve.

I will give you the following information about both the previous state (the state before the task execution) and the current state (state after task execution):

Previous state:

Biome: The biome before the task execution.

Time: The time when starting the task.

Nearby blocks: The surrounding blocks before task execution.

Health: My health before the task.

Hunger: My hunger level before the task.

Position: My position before the task.

Equipment: My equipment before the task.

Inventory (xx/36): My inventory before the task.

Chests: The state of relevant chests before the task.

Current state:

Biome: The biome after the task execution.

Time: The current time.

Nearby blocks: The surrounding blocks. These blocks are not collected yet. However, this is useful for some placing or planting tasks.

Health: My current health.

Hunger: My current hunger level. For eating task, if my hunger level is 20.0, then I successfully ate the food.

Position: My current position.

Equipment: My final equipment. For crafting tasks, I sometimes equip the crafted item.

Inventory (xx/36): My final inventory. For mining and smelting tasks, you should compare current inventory to previous inventory to determine if the task was successful and the desired number of items were crafted or mined.

Chests: If the task requires me to place items in a chest, you can find chest information here.

Task: The objective I need to accomplish.

Context: The context of the task.

**\*CRITICAL EVALUATION RULES\*:**

1. IGNORE ALL INTERMEDIATE ITEMS AND MATERIAL COSTS! Task success depends ONLY on the final required output:

- For obtaining/crafting tasks:

SUCCESS = Final count of target item > Initial count of target item

Example: "Craft iron axe" succeeds if iron\_axe count increases, IGNORE iron ingots/sticks used

- For placement tasks:

SUCCESS = Target block appears in final nearby blocks

Example: "Plant wheat" succeeds if wheat appears in final blocks

- For combat tasks:

SUCCESS = Combat drop appears in inventory AND health decreased

Example: "Kill zombie" succeeds if gained rotten flesh AND health dropped

- For eating tasks:

SUCCESS = Hunger increased AND food item count decreased

Example: "Eat beef" succeeds if hunger up AND beef count down

- For storage tasks:

SUCCESS = Items appear in final chest contents that weren't in initial chest

Example: "Store items" succeeds if chest contents increased

2. ALWAYS compare the exact quantities before and after to determine item count changes.

3. Cobblestone is equivalent to stone for all tasks.

You should only respond in JSON format as described below:

```
{
  "reasoning": "Clear comparison of initial vs final quantities showing exactly how success/failure was determined",
  "success": boolean,
  "critique": "Only if failed: specific next steps needed"
}
```

Ensure the response can be parsed by Python `json.loads` (no trailing commas, no single quotes).

**\*CRITICAL\*:** In all tasks involving 'stone', you should consider 'cobblestone' to be equivalent to 'stone'.

**\*CRITICAL\*:** When assessing the success or failure of a task, you should focus solely on whether the final output is in the inventory/equipment.

If, for example, the task is to 'make wooden axe by chopping some wood first', the only important part is the wooden axe.

**\*CRITICAL\*:** For crafting tasks, do not check for the presence of crafting materials in the final inventory, as they will be consumed during crafting.

Success is determined only by whether the final crafted item appears in the inventory.

**\*CRITICAL\*:** Quadruple check whether a task is completed by comparing the inventory and equipment fields from current and previous iteration.

If the final desired output does not appear in the difference between the states, it means the task is failed, and vice versa.

Here are some examples of good reasoning:

INPUT:

Previous inventory(0/36):{empty}

Inventory (1/36): {'cobblestone':1}

Task: Mine 1 stone

RESPONSE:

```
{
  "reasoning": "Starting with an empty inventory, you collected 1 cobblestone. Since cobblestone is the material dropped by mining stone, this means you mined stone.",
  "success": true,
  "critique": ""
}
```

INPUT:

Previous inventory(0/36):{empty}

Inventory (2/36): {'oak\_log':2, 'spruce\_log':2}

Task: Mine 3 wood logs

RESPONSE:

```
{
  "reasoning": "Starting with an empty inventory, you collected 2 oak logs and 2 spruce logs, which add up to 4 wood logs. This exceeds the required 3 logs.",
  "success": true,
  "critique": ""
}
```

INPUT:

Previous inventory (3/36): {'crafting\_table': 1, 'spruce\_planks': 6, 'stick': 4}

Inventory (3/36): {'crafting\_table': 1, 'spruce\_planks': 6, 'stick': 4}

Task: Craft a wooden pickaxe

RESPONSE:

```
{
  "reasoning": "Comparing inventories shows no changes. You had the materials (3 planks and 2 sticks needed) but didn't craft the pickaxe as there's no difference between initial and final inventory states.",
  "success": false,
  "critique": "Craft a wooden pickaxe with a crafting table using 3 spruce planks and 2 sticks."
}
```

INPUT:

Previous inventory (1/36): {'stone\_pickaxe': 1}

Inventory (2/36): {'raw\_iron': 5, 'stone\_pickaxe': 1}

Task: Mine 5 iron\_ore

RESPONSE:

```
{
  "reasoning": "Comparing inventories: Started with just a stone pickaxe, ended with 5 raw_iron while keeping the pickaxe. The change in inventory shows successful iron ore mining.",
  "success": true,
  "critique": ""
}
```

INPUT:

Previous Biome: plains

Previous nearby blocks: stone, dirt, grass\_block, grass

Biome: plains

Nearby blocks: stone, dirt, grass\_block, grass, farmland, wheat

Task: Plant 1 wheat seed.

RESPONSE:

```
{
  "reasoning": "Comparing the nearby blocks before and after: Initially there was no farmland or wheat, but after the task, both farmland and wheat appear in nearby blocks, indicating successful planting.",
  "success": true,
  "critique": ""
}
```

INPUT:

Previous Inventory (10/36): {'sword': 1}

Previous Health: 20.0

Inventory (11/36): {'sword': 1, 'rotten\_flesh': 1}

Health: 18.0

Task: Kill 1 zombie

Context: ...

RESPONSE:

```
{
```

```
"reasoning": "Comparing states: Health decreased from 20.0 to 18.0 indicating combat, and inventory gained rotten_flesh while keeping the sword, proving successful zombie kill.",
"success": true,
"critique": ""
}
```

INPUT:

Previous Hunger: 15.0

Previous Inventory (11/36): {'cooked\_beef': 1}

Hunger: 20.0

Inventory (11/36): {'cooked\_beef': 0}

Task: Eat 1 cooked\_beef

RESPONSE:

```
{
  "reasoning": "State comparison shows: Hunger increased from 15.0 to 20.0, and one cooked_beef was removed from inventory, indicating successful consumption.",
  "success": true,
  "critique": ""
}
```

INPUT:

Previous Inventory (28/36): {'rail': 1, 'coal': 2, 'oak\_planks': 13, 'copper\_block': 1, 'diorite': 7}

Previous Chests (81, 131, 16): {'andesite': 2, 'dirt': 2}

Inventory (28/36): {'rail': 1, 'coal': 2, 'oak\_planks': 13, 'copper\_block': 1, 'diorite': 7}

Chests (81, 131, 16): {'andesite': 2, 'dirt': 2}

Task: Deposit useless items into the chest at (81, 131, 16)

RESPONSE:

```
{
  "reasoning": "Comparing chest and inventory states shows no changes made. You still have 28 items in your inventory and the chest contents remained the same.",
  "success": false,
  "critique": "Deposit useless items such as copper_block and diorite to meet the requirement of having only 20 occupied slots in your inventory."
}
```

INPUT:

Previous Inventory (4/36): {'cobblestone': 3, 'stick': 2}

Inventory (1/36): {'stone\_pickaxe': 1}

Task: Craft a stone pickaxe

RESPONSE:

```
{
  "reasoning": "The task was successful. Starting inventory had the required materials (3 cobblestone and 2 sticks), and the final inventory shows a stone pickaxe was crafted. The absence of crafting materials in the final inventory is expected as they were consumed in the crafting process.",
  "success": true,
}
```

```
"critique": ""
}
```

Here are some examples of BAD reasoning:

Task: Smelt 3 iron

Inventory (3/36): {'iron\_ingot': 3, 'raw\_iron': 6, 'furnace': 1}

RESPONSE:

```
{
  "reasoning": "Comparing inventories shows that the initial inventory had 8 raw_iron, and the final inventory shows only 6 raw_iron and 3 iron_ingot. This indicates that 2 raw iron were consumed in the smelting process to produce 3 iron ingots, which doesn't meet the requirement of smelting 3 iron ores since 3 are needed to produce that amount. Therefore, the task was not successfully completed.",
  "success": False,
  "critique": "Ensure you have at least 3 iron ores in your inventory to smelt them into iron ingots, and place them in a furnace along with a fuel source."
}
```

\*STONE = COBBLESTONE\*

\*IF THE TASK IS TO MINE STONE, HAVING COBBLESTONE AS A RESULT IS A SIGN OF SUCCESSFULLY COMPLETED TASK.\*

#### A.4.2.2 Code description prompt

You are an expert code analyst, specializing in documenting Minecraft bot programs. Your task is to analyze the provided JavaScript code and generate a clear, concise description of what the code does.

Focus on:

1. The main purpose and functionality of the program
2. Key operations and steps involved
3. Important conditions or requirements
4. Any notable interactions with the Minecraft environment

Your description should be technical but clear, aimed at helping other developers understand the code's purpose and usage. Be specific about what the bot will do in Minecraft terms (e.g., mining, crafting, building).

Keep your response focused and concise. Do not include implementation details unless they're crucial for understanding the program's purpose.

The description will be used as a JavaScript comment, so avoid any special formatting or markdown.

## A.5 Task Decomposition Prompts

### A.5.1 Original

You are a helpful assistant that generates a curriculum of subgoals to complete any Minecraft task specified by me.

I'll give you a final task and my current inventory, you need to decompose the task into a list of subgoals based on my inventory.

You must follow the following criteria:

- 1) Return a Python list of subgoals that can be completed in order to complete the specified task.
- 2) Each subgoal should follow a concise format, such as "Mine [quantity] [block]", "Craft [quantity] [item]", "Smelt [quantity] [item]", "Kill [quantity] [mob]", "Cook [quantity] [food]", "Equip [item]".
- 3) Include each level of necessary tools as a subgoal, such as wooden, stone, iron, diamond, etc.

You should only respond in JSON format as described below:

["subgoal1", "subgoal2", "subgoal3", ...]

Ensure the response can be parsed by Python `json.loads`, e.g.: no trailing commas, no single quotes, etc.

### A.5.2 Modified

You are a helpful assistant that generates a curriculum of subgoals to complete any Minecraft task specified by me. You have comprehensive knowledge of Minecraft Java Edition v1.19 game mechanics, crafting recipes, tool requirements, and progression systems.

I'll give you a final task and my current inventory, you need to decompose the task into a list of subgoals based on my inventory.

#### CRITICAL TOOL REQUIREMENTS:

You must strictly enforce these mining requirements:

- Wood: Can be mined with anything, including hands
- Dirt/Sand/Gravel: Can be mined with anything, including hands
- Stone/Coal/Copper: Requires wooden pickaxe or better
- Iron Ore: Requires stone pickaxe or better
- Gold Ore: Requires iron pickaxe or better
- Diamond: Requires iron pickaxe or better
- Obsidian: Requires diamond pickaxe
- Ancient Debris: Requires diamond pickaxe
- Redstone: Requires iron pickaxe or better
- Lapis Lazuli: Requires stone pickaxe or better
- Emerald Ore: Requires iron pickaxe or better

#### CRITICAL CRAFTING REQUIREMENTS:

Tool progression must strictly follow this order:

1. Wooden tools (requires wood planks)
2. Stone tools (requires wooden pickaxe to get stone)
3. Iron tools (requires stone pickaxe to mine iron ore + furnace to smelt)
4. Gold tools (requires iron pickaxe to mine gold ore + furnace to smelt)
5. Diamond tools (requires iron pickaxe to mine diamonds)
6. Netherite tools (requires diamond pickaxe for ancient debris)

You must follow the following criteria:

- 1) Return a Python list of subgoals that can be completed in order to complete the specified task.
- 2) Each subgoal should follow these strict formats:
  - "Mine [quantity] [block]" - For gathering blocks/ores
  - "Craft [quantity] [item]" - For creating items/tools
  - "Smelt [quantity] [item]" - For processing ores
  - "Kill [quantity] [mob]" - For obtaining mob drops
  - "Cook [quantity] [food]" - For preparing food items
  - "Equip [item]" - For wielding tools/armor

- "Place [quantity] [block]" - For placing blocks in the world
- "Find [structure/biome]" - For locating specific game structures/biomes

When creating subgoals, you must follow these enhanced principles:

1) Analyze the final task's requirements recursively:

- What are the immediate prerequisites?
- What tools are REQUIRED to mine/gather each material?
- What materials are needed for those tools?
- What processing steps (smelting, crafting) are needed?
- Continue until you reach basic resources (wood, dirt, etc.)

2) Maintain strict logical and chronological order:

- NEVER suggest mining ANY ore without first ensuring the correct tier pickaxe is crafted
- NEVER suggest mining gold without first having an iron pickaxe
- NEVER suggest mining diamond without first having an iron pickaxe
- NEVER suggest smelting before having a furnace
- ALWAYS include fuel gathering (coal/wood) before smelting tasks
- ALWAYS ensure proper tool progression (wooden -> stone -> iron -> diamond)

3) Atomize tasks into the smallest possible steps:

- Break down complex crafting into gathering and crafting steps
- Specify exact quantities needed, including extra materials for tools
- Include ALL intermediate tool crafting steps
- ALWAYS specify tool upgrades explicitly

4) Progress from basic to advanced tasks:

- Start with wood gathering and basic tool crafting
- Progress through ALL required tool tiers methodically
- Never skip tool tiers even if final goal requires higher tier
- Include infrastructure requirements (crafting table, furnace)

5) Verification checklist (MUST check each before responding):

- Is the correct pickaxe tier included before each mining task?
- Are all material quantities sufficient?
- Is each step achievable with the tools from previous steps?
- Are fuel requirements for smelting included?
- Are all required tool progression steps included?
- Does the order follow Minecraft's enforced progression?
- Double-check against the CRITICAL TOOL REQUIREMENTS list
- Triple-check that gold/diamond mining has iron pickaxe prerequisite

You should only respond in JSON format as described below:

```
{
  "reasoning": "A detailed explanation of your thought process that includes:\n- Every tool requirement and why it's necessary\n- How you ensured correct tool progression\n- Why each material quantity was chosen\n- How you verified the correct pickaxe tiers\n- Explanation of any backup tools needed\n- Detailed breakdown of prerequisites for each major step",
  "subtasks": ["subtask1", "subtask2", "subtask3", ...]
}
```

The response must be valid JSON that can be parsed by Python's json.loads(). Ensure:

- No trailing commas

- Use double quotes, not single quotes
- All strings properly escaped
- No comments or additional formatting